From *Moringa oleifera* Radiobiology to Explainable Machine Learning

Cellular Radiosensitivity Prediction and Evidence-Based Compound Prioritisation

A reproducible investigation linking an experimental study of *Moringa oleifera* seed and leaf extracts in irradiated cells to an explainable benchmark of clonogenic radiosensitivity across 237 independent cancer cell-line components, and to a transparent, evidence-based shortlist of plant constituents for future experimental validation.

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Document information

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Not executed	Negative controls (1,000 draws of each null) and the coverage simulation for an interval method (Chapter 10)
Outcome data licence	CLARIFi v1.0, CC BY-NC 4.0; data DOI `10.5281/zenodo.20710748` and the licence must be cited in any reuse

Every number in this report is taken from a saved output of the project pipeline; the source file of each result table is named in its caption, and the correspondence with the project's table identifiers is given in Table S9. The figures are the vector graphics produced by the project's figure scripts. No model was refitted for this report.

Executive summary

Research motivation. An experimental study of crude dichloromethane (DCM) seed and leaf extracts of *Moringa oleifera* Lam. observed that the extracts changed how irradiated cells proliferated, and that the change differed between an apparently normal rodent fibroblast line (V79) and two human carcinoma lines (DU145 and MCF-7). A three-line experiment cannot say whether such differences reflect the cells' *intrinsic* radiation response, nor whether measurable molecular features explain it. This project asked that broader question on a defensible endpoint and with public data, and then returned to the plant to ask which of its documented constituents deserve experimental testing.

Computational question. Does baseline gene expression, summarised as 50 Hallmark pathway scores, predict experimentally measured clonogenic radiosensitivity across cancer cell lines better than prespecified simple baselines, once related cell lines are kept together and tissue of origin is already taken into account?

Data and design. We assembled 241 eligible cell-line records, forming 237 independent biological components, by linking literature-derived clonogenic surviving fractions at 2 Gy (CLARIFi mSF2) to baseline expression (DepMap 24Q4). Cell-line identity and relatedness were resolved through Cellosaurus, and pathway features were computed from the MSigDB Hallmark gene sets. Five prespecified models, from a global mean to tissue, proliferation and expression together, were compared under ten repeats of five-fold cross-validation grouped by biological component and stratified by tissue, with every tuning and scaling step confined to training folds. All 250 model-fold tasks completed.

Principal findings. Adding expression features to a tissue-only baseline reduced the mean absolute error (MAE) by **0.00383 mSF2 units, 2.85% of the baseline error**, and the richer model had the lower error in **nine of ten** cross-validation repeats. Out-of-sample R^2 stayed close to zero for every model (-0.018 to $+0.013$). Expression therefore carries a small amount of usable information beyond tissue of origin, while overall predictive power is weak — a result that agrees with the strongest published benchmark in the field. Superiority was not formally tested: a protocol deviation, recorded before any result was seen, deferred the coverage simulation needed to validate an interval method, and no negative-control draws were run. The improvement is reported as a descriptive estimate.

Robustness and generalisation. Three further analyses of the same frozen cohort tested how durable this small advantage is. A random forest (M3) did **worse** than the linear offset model (gain over the tissue baseline 0.00275 against 0.00383). On transformed outcome scales the advantage kept its direction but almost vanished on the neglog scale (0.00023 in neglog units) and was larger in absolute terms on the logit scale (0.00980 in logit units), so its size depends on how the outcome is expressed. When one tissue at a time was held out, an expression-only model reached out-of-sample R^2 of **+0.031** against a global mean, the only clearly positive R^2 in the project; it nevertheless lost to the global mean

when Pancreas (−0.0142) or Esophagus/Stomach (−0.0030) was held out, so transfer to an unseen tissue is conditional on the tissue.

Explainability. Held-out permutation importance ranked the 50 Hallmark scores. Seven reached the prespecified exploratory tier C, led by the p53 pathway; the higher tiers were unreachable by design, because they require a formally demonstrated prediction gain and at least 1,000 negative-control draws. Importance magnitudes are as small as the predictive gain itself, rankings are only moderately stable across fits, and the DNA repair, G2M checkpoint and hypoxia sets that radiobiology would lead one to expect were not prioritised.

Compound prioritisation. *M. oleifera* seed and leaf constituents were curated from primary analytical studies, with the extraction solvent of every report recorded, and linked to radiation-response biology through a four-dimension score of 0–8 points fixed before any score was computed. Eighteen compounds were ranked and five reached tier A. The curation exposed the central mismatch of the project: **niazimicin is the only constituent with a structurally confirmed identification in a dichloromethane fraction of both seed and leaf, yet no study has tested the purified compound in any radiation-response process; conversely, the compounds with the strongest radiation evidence have never been found in a dichloromethane extract.** Two prespecified safeguards visibly bound: chlorogenic acid scored 7 of 8 but is held at tier C because one study reports that it protects *tumour* cells from radiation, and benzyl isothiocyanate is excluded despite strong radiation evidence because it is a root, not a seed or leaf, constituent. A tier summarises the state of the published evidence; it never asserts radioprotection.

Significance. The project delivers a reproducible, auditable pipeline, an honest benchmark whose negative-leaning result agrees with the best prior work, and a traceable route from the original radiobiology to a short list of experiments that could settle the question. Its value lies in the rigour of its methods and the transparency of its evidence rather than in a strong performance claim.

> **Recommended next steps.** Characterise the original DCM extracts analytically (niazimicin and moringin first); complete the deferred coverage simulation and the negative controls; and test α-tocopherol and β-sitosterol, purified and against whole extract, on clonogenic survival in matched normal and tumour cells over a dose range.

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1 | Introduction and scientific background

1.1 Cellular responses to ionising radiation

Ionising radiation kills cells mainly by damaging DNA, and the DNA double-strand break is the lesion most closely tied to loss of reproductive capacity. The reference measure of how much radiation a cell population survives is the *clonogenic assay*, which counts the cells that retain unlimited proliferative capacity and form colonies after irradiation (**puck1956**). For a dose D, the surviving fraction is the plating efficiency of irradiated cells relative to that of unirradiated controls,

$$\mathrm{SF}(D) \;=\; \frac{N_{\mathrm{colonies}}(D) \,/\, N_{\mathrm{seeded}}(D)}{N_{\mathrm{colonies}}(0) \,/\, N_{\mathrm{seeded}}(0)}. \tag{1.1}$$

The surviving fraction at 2 Gy (SF2) is the most widely used single summary, because 2 Gy is the conventional daily fraction in radiotherapy and because SF2 has been associated with clinical outcome in some settings (**fertil1981**; **west1993**).

Cell lines differ substantially and reproducibly in intrinsic radiosensitivity, and these differences are only partly explained by tissue of origin (**yard2016**). The determinants include DNA damage response and repair capacity, cell-cycle checkpoint control, proliferation rate, apoptotic propensity, oxygenation and redox state. Each of these processes leaves a transcriptional footprint, which is why prediction from gene expression is plausible in principle.

1.2 The Moringa oleifera experimental motivation

The motivating study is an unpublished manuscript, ‘Dichloromethane seed and leaf extracts of *Moringa oleifera* (Lam) enhance proliferation in irradiated rodent fibroblasts *in vitro*’. The scanned copy available to the project is incomplete: printed pages 1, 2, 4, 5, 6, 13, 14 and 15 are present, while page 3 and pages 7–12 are missing. The plant material and extraction methods, the end of the discussion, the conclusions and the reference list therefore cannot be read; authors, venue, date and DOI are not shown, and no published version was located. No missing method, result or conclusion has been reconstructed in this report.

The available pages document the following design. Crude DCM seed and leaf extracts were reconstituted in phosphate-buffered saline, sterile-filtered and applied at 0–100 µg/mL to V79 (described as apparently normal Chinese hamster lung fibroblasts), DU145 (human prostate cancer) and MCF-7 (human mammary adenocarcinoma). For each cell line and extract, one plate was left unirradiated and a second received 2 Gy of X-rays at 1 Gy/min. Adherent biomass was measured 72 h later by crystal-violet optical density at 590 nm, with dose-matched untreated controls subtracted. The authors integrated the change in optical density over extract concentration, giving an area A_D at each dose D, and reported a *proliferation enhancement factor*

$$\mathrm{PEF} \;=\; \frac{A_{2\,\mathrm{Gy}}}{A_{0\,\mathrm{Gy}}}. \tag{1.2}$$

For the seed extract, PEF exceeded 1 in all three lines (V79 1.25, DU145 1.12, MCF-7 1.16). For the leaf extract, PEF was 1.26 in V79, 0.90 in DU145 and 0.99 in MCF-7.

Four cautions follow directly from the design, and they constrain everything that follows.

1. **PEF is not SF2.** PEF is a ratio of integrated extract-versus-control differences in 72-hour biomass measured with the extract present; SF2 is a surviving fraction of untreated cells measured over one to three weeks of colony growth. The two are never equated, pooled or plotted on one axis in this project.
2. **'Enhancement' denotes two different quantities in the manuscript.** By PEF the seed extract enhanced all three lines; by growth relative to untreated control only V79 increased, while the two carcinoma lines showed no change or reduced growth. The discussion's statement about the fibroblasts refers to the second quantity, the results' statement about all lines to the first.
3. **The normal-versus-malignant comparison is confounded.** The only apparently normal line is also the only non-human line, so species, tissue of origin, phenotype and malignancy vary together. Only the leaf extract shows a normal/malignant difference in PEF.
4. **The design cannot support a selectivity claim.** There is a single non-zero dose and no dose–response; the replicates are technical wells rather than independent biological repeats; dose is confounded with plate; and no vehicle or cell-free interference control is described. The available pages contain no clonogenic, cytotoxicity, apoptosis, DNA-damage or migration assay, no normal human cell line and no mechanistic experiment.

The *Moringa* measurements are therefore used as *biological motivation only*. They are never used as training, calibration or validation data, and no claim of selective phytochemical radioprotection is made anywhere in this report.

1.3 Why cellular radiosensitivity is hard to predict

Published attempts to predict cell-line radiosensitivity from expression have performed poorly under honest evaluation. The most directly relevant prior work, RadSigBench, re-fitted seven published radiosensitivity signatures and benchmarked them against resampled random gene sets, reporting a mean out-of-sample R^2 of 0.01 on CCLE data and 0.00 on NCI-60, with the published signatures performing no better than random gene sets (**oconnor2022**). This is the single most important prior for the present work: it sets the expectation that the signal is weak, so a near-zero R^2 here is a replication rather than a surprise.

Three difficulties recur. The first is *sample size*: panels with reliable clonogenic data number in the hundreds, against tens of thousands of candidate molecular features. The second is *label noise*: literature-aggregated SF2 values combine laboratories, dose rates and protocols, and an audit of 1,672 radiobiology papers found that 30.5% did not report replicates and 32.3% did not report dose rate (**oike2020**). The third is *leakage through relatedness*: isogenic derivatives, sublines and cross-contaminated lines share molecular profiles, so a split that separates records rather than biological entities inflates apparent performance.

1.4 The case for public radiogenomic data

A three-line extract experiment cannot address intrinsic radiosensitivity across cell types, but public resources can. CLARIFi aggregates literature clonogenic radiosensitivity (**clarifi2024**); DepMap provides harmonised baseline expression for the same lines (**depmap24q4**); Cellosaurus provides authoritative identity, problem flags and relatedness (**bairoch2018**); and the MSigDB Hallmark collection provides 50 compact, interpretable pathway summaries (**liberzon2015**). Together they allow a properly grouped, prespecified benchmark on a defensible endpoint.

1.5 The value and the limits of interpretable machine learning

Interpretable models matter here for two reasons: a pathway-level summary is biologically meaningful, and feature attributions can be examined for stability. Attribution, however, has hard limits. Permutation importance describes a *fitted model*, not biology; correlated features share or exchange importance; and the size of an importance is not a molecular effect size. Attribution is only as trustworthy as the prediction it explains, which is why every pathway statement in this report is capped at an exploratory tier while the prediction gain remains small and unvalidated.

1.6 Rationale for prioritising candidate compounds

The original study raises a question that the machine-learning benchmark alone cannot answer: if cells differ in radiation response, and some of that difference is reflected in pathway activity, which *Moringa* constituents are worth testing? The answer has to be built compound by compound from independently published chemistry and biology, with the extraction solvent tracked explicitly, because the extracts used in the original study were never chemically characterised.

2 | Aim and objectives

Aim. To build a reproducible, auditable research pipeline that connects the original *Moringa oleifera* irradiation study to a transparent, explainable benchmark of cellular radiosensitivity prediction on public radiogenomic data, and to use that benchmark, together with curated published evidence, to produce a traceable and explicitly exploratory shortlist of candidate constituents and a feasible strategy for their experimental validation.

Figure 1 shows the six stages of the project and the order in which they were carried out. Stages 1–5 are complete; stage 6 is a proposal.

Table 1 lists the ten objectives and their status. Nine are complete. The validation of the primary comparison by a coverage simulation and negative controls (objective 9) was deferred by a protocol deviation recorded before any result existed (Section 10.2); three other robustness analyses were executed in its place (Chapter 6).

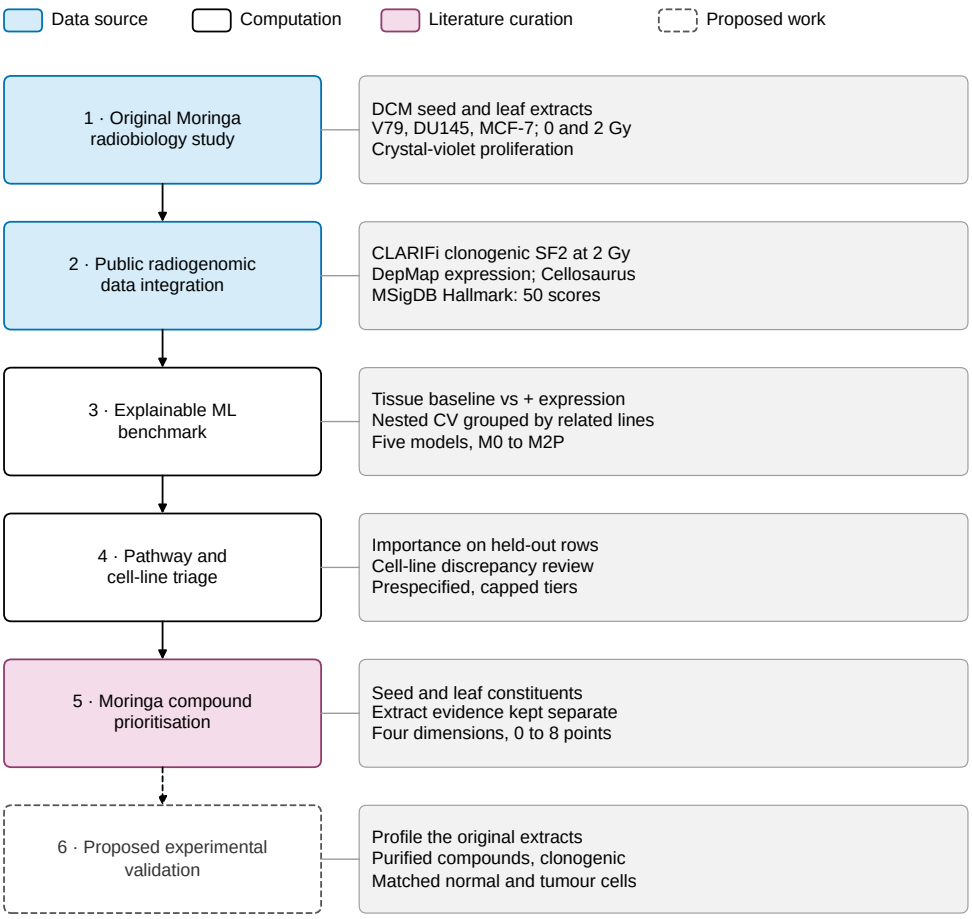


Figure 1. Structure of the project, from the original *Moringa oleifera* radiobiology investigation to the prioritisation of candidate constituents for future experimental work. Box fills distinguish data sources, computational steps, literature curation and proposed future work; they do not indicate the strength of the evidence at each stage. The *Moringa* measurements are biological motivation only and are never used as training, calibration or validation data, because crystal-violet proliferation under extract treatment and clonogenic survival at 2 Gy are different endpoints. Stage 6 is a proposal, not completed work.

Table 1. Project objectives and their status.

#	Objective	Status
1	Extract and document the original manuscript's methods and results, marking every unreadable or absent item	Completed (Section 1.2)
2	Assemble a provenance-tracked dataset linking clonogenic SF2 to baseline expression, with cell-line identity and relatedness resolved	Completed; 241 records, 237 components
3	Prespecify and execute a grouped, nested cross-validated comparison of five models on a single named endpoint	Completed; 250/250 model-fold tasks
4	Quantify the predictive gain of expression beyond tissue of origin and report it with its limitations	Completed; descriptive estimate
5	Rank Hallmark gene-set scores by held-out permutation importance under prespecified evidence gates	Completed; capped at tier C
6	Triage the cell lines whose measured radiosensitivity the model reproduces least well, separating technical from biological explanations	Completed
7	Curate *M. oleifera* seed and leaf constituents and their radiation-response evidence from primary sources, tracking the extraction solvent	Completed (Tables S5 and S6)
8	Apply a transparent 0–8 point prioritisation fixed before computation and produce the ranked shortlist	Completed; 18 ranked, 5 in tier A
9	Validate the comparison with a coverage simulation and negative controls	Deferred by recorded protocol deviation
10	Specify a feasible experimental validation strategy	Completed as a proposal (Chapter 12)

3 Data sources and resources

The resources used by the project are listed in Table 2. Every raw file carries a manifest recording its source, DOI, URL, access date, retrieval method, SHA-256 checksum, size, column meanings, units, endpoint definition and transformations; the data loaders verify the checksum and stop on any mismatch. The outcome file, the expression matrix, the Hallmark gene-set file and the quality-control outputs were all verified against their pinned checksums before the analysis run.

Licensing and provenance caveats.

- **CLARIFi is licensed CC BY-NC 4.0** (non-commercial), as confirmed from both the Crossref licence record and the Europe PMC full text. The raw file is kept outside version control, the data DOI and the licence must be cited in any publication, and the non-commercial terms constrain downstream reuse.
- The **Hallmark gene-set file** was downloaded manually, so no acquisition log or published checksum exists for it. Its content was verified locally (48,689 bytes, 50 unique `HALLMARK_` sets, standard layout, SHA-256 `f22066af…cbbdd8e9`), but its identity with the official release has not yet been confirmed against the official download address.
- The **original manuscript may not be redistributed**; only its checksum and the page-by-page extraction record are kept.
- Compound curation used public databases and published literature only; no licensed chemical database was queried.

Table 2. Data resources, releases, identifiers and licences.

Resource	Release	Identifier	Licence	Role in the project
Original *Moringa* manuscript	Unpublished scan, 8 pages	SHA-256 `7f644f55…e870fb9`	Redistribution not authorised	Biological motivation only
CLARIFi radiosensitivity	v1.0	Data `10.5281/zenodo.20710748`; paper `10.1111/cas.16334`	CC BY-NC 4.0	Outcome: mSF2
DepMap	24Q4 Public	`10.25452/figshare.plus.27993248.v1`	CC BY 4.0	Baseline expression; cell-line metadata
Cellosaurus	r56.0	`10.26037/yareta:lbz6rhlaavbldmavv4k5jdaphe`	CC BY 4.0	Identity, problem flags, relatedness
MSigDB Hallmark	2025.1.Hs	Paper `10.1016/j.cels.2015.12.004`	MSigDB terms	50 pathway features
PubChem, Europe PMC, Crossref	Accessed 22 Oct 2026	—	Public interfaces	Compound identity; literature verification

4 | Methodology

4.1 The original experimental record

The *Moringa* manuscript was read page by page, and every extracted item was labelled as a measured result, an authors' interpretation, this project's commentary, or not stated; nothing absent was reconstructed. A radiobiology review checked every extracted item against the page images. Internal inconsistencies were recorded as printed rather than silently corrected: the extract concentration appears as 25–50 µg/mL in the abstract, 30–50 mg/mL in the results and above 20 µg/mL in the discussion; V79 is described variously as hamster, murine and rodent; and the tabulated integrated areas are all positive although the plotted optical-density changes are mostly negative for the two carcinoma lines, which cannot be reconciled without the missing pages or the raw data.

4.2 Data acquisition, harmonisation and eligibility

Outcome, expression, identity and pathway resources were acquired under recorded provenance, parsed into separate per-endpoint tables and harmonised on DepMap identifiers. Download code was written only after each source had been verified; raw files are immutable, no value was invented or silently imputed, and missingness is documented.

Cell lines were matched by stable identifiers, never by name. Cellosaurus supplied accessions, synonyms, problem flags, and the hierarchy and origin links used to detect relatedness. A record was eligible if it had a usable clonogenic SF2 value, a matched baseline expression profile and a resolved identity, which gave **241 eligible records** from 20 tissues.

4.3 Endpoint and features

The single primary endpoint is **CLARIFi mSF2**, the median of the literature-reported clonogenic surviving fractions at 2 Gy for a cell line, analysed on its raw scale. Predictions are clipped to $[0, 1]$ by a prespecified rule; outcomes are never clipped. Crystal-violet proliferation, cell viability, ATP-based integral survival and differential expression are kept strictly separate from clonogenic SF2 throughout the project.

Three kinds of feature were used: tissue of origin (OncotreeLineage, one-hot encoded over fixed categories), an optional proliferation covariate, and 50 Hallmark gene-set scores computed from baseline expression (DepMap $\log_2(\mathrm{TPM}+1)$ values of protein-coding genes). Within each training fold, every gene g is standardised with the mean μ_g and standard deviation σ_g of the training rows only, and the score of set S for cell line i is the mean standardised expression of its member genes present in the matrix,

$$s_{iS} = \frac{1}{|S^*|} \sum_{g \in S^*} \frac{x_{ig} - \mu_g}{\sigma_g}, \qquad S^* = S \cap \{\text{measured genes}\}. \tag{4.1}$$

The proliferation covariate is the combined score of the E2F-targets and G2M-checkpoint sets.

Table 3. The five prespecified models of the primary benchmark.

Model	Features	Role
M0	None (training-set mean)	Floor
M1	Tissue of origin	Primary comparator
M1P	Tissue and proliferation score	Sensitivity comparator
M2	**Tissue and 50 Hallmark expression scores**	**Primary model**
M2P	Tissue, proliferation and the other 48 Hallmark scores	Sensitivity model

4.4 Grouping and cross-validation

Records of cell lines related by identity (isogenic derivatives, sublines, documented relatedness) were merged into **237 independent biological components**; seven components contain more than one record. Every split is grouped by component, so related lines can never fall on both sides of a train/test boundary. Splits are stratified by tissue: the nine tissues with at least ten components each form their own stratum and the remaining eleven are pooled.

Validation used **ten repeats of five-fold cross-validation** with an inner five-fold loop for hyperparameter selection. All preprocessing, scaling, gene-set scoring and tuning took place **inside training folds only**. Each record i belonging to component $c(i)$ carries the weight

$$w_i \;=\; \frac{1}{n_{c(i)}}, \tag{4.2}$$

where n_c is the number of records of component c, in every fit and every inner score, so that a component with several records counts once. The base random seed (20261021) and all grids are recorded in the experiment configuration. A dedicated leakage audit covered identifier handling, component construction and split construction.

4.5 Models

Five models were prespecified, from simple to rich (Table 3), and simple scikit-learn estimators were used before any more complex architecture. M1 is a ridge regression on the fixed tissue categories with an unpenalised intercept, which shrinks each tissue mean towards the overall mean. M2 is an *offset model*: within each training fold it first fits M1 and then an elastic net to the M1 residuals using the standardised gene-set scores $\tilde{\mathbf{s}}_i$,

$$\hat{y}_i^{\mathrm{M2}} \;=\; \hat{y}_i^{\mathrm{M1}} + \beta_0 + \tilde{\mathbf{s}}_i^{\top}\beta, \qquad \hat{\beta} = \arg\min_{\beta_0,\beta} \frac{\sum_i w_i (r_i - \beta_0 - \tilde{\mathbf{s}}_i^{\top}\beta)^2}{2\sum_i w_i} + \alpha\rho\,\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2, \tag{4.3}$$

with $r_i = y_i - \hat{y}_i^{\mathrm{M1}}$. Because M2 contains M1 as an offset, the comparison of the two asks specifically what expression adds beyond tissue. M1P and M2P are the corresponding models with the proliferation covariate added to the baseline; M2P uses the remaining 48 Hallmark scores in its residual model. The inner loop searched $\alpha \in \{0.1, 0.01, 0.001\}$, $\rho \in \{0.9, 0.5, 0.1\}$ and the baseline ridge penalty $\in \{100, 10, 1, 0.1\}$, 36 combinations for M2.

Two further models were configured for the exploratory layer (Chapter 6): M2X, an elastic net on the 50 scores with the global mean as offset and no tissue information, used only for the leave-one-tissue-out analysis; and M3, a random forest on one-hot tissue and the 50 scores. M3 was added after the primary results had been seen and therefore cannot replace M2 as the primary benchmark, whatever its score.

4.6 Performance measures and the primary estimand

For a repeat r with out-of-fold predictions $\hat{y}_{ir}$, the component-weighted error measures are

$$\mathrm{MAE}_r = \frac{\sum_i w_i |y_i - \hat{y}_{ir}|}{\sum_i w_i}, \qquad \mathrm{RMSE}_r = \left(\frac{\sum_i w_i (y_i - \hat{y}_{ir})^2}{\sum_i w_i} \right)^{1/2}, \tag{4.4}$$

$$R^2_{\mathrm{oos},r} = 1 - \frac{\sum_i w_i (y_i - \hat{y}_{ir})^2}{\sum_i w_i (y_i - \bar{y}_w)^2}, \qquad \bar{y}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}, \tag{4.5}$$

and each reported value is the mean over the $R = 10$ repeats. $R^2_{\mathrm{oos}} = 0$ therefore corresponds to predicting the cohort mean, and negative values are worse than that. Calibration slope and intercept come from a weighted regression of observed on predicted values.

The primary estimand is the paired, component-weighted difference in mean absolute error between M1 and M2. With the record-level difference $d_{ir} = |y_i - \hat{y}^{\mathrm{M1}}_{ir}| - |y_i - \hat{y}^{\mathrm{M2}}_{ir}|$, the component means $D_{cr} = n_c^{-1} \sum_{i \in c} d_{ir}$ and the $C = 237$ components,

$$\Delta\mathrm{MAE} = \frac{1}{R} \sum_{r=1}^{R} \frac{1}{C} \sum_{c=1}^{C} D_{cr}, \tag{4.6}$$

so a positive value favours the richer model. MAE, RMSE and R^2_{oos} are reported for every model, and no metric was chosen after the results were seen.

4.7 Explainability

For each of the 50 real-data fits of M2 (ten repeats by five folds), grouped permutation importance was computed on the *outer-test rows only*, using the saved model, with 20 permutations per feature; nothing was refitted. For gene set S in fit f,

$$I_{S,f} = \frac{1}{20} \sum_{p=1}^{20} \left[\mathrm{MAE}_f(\text{score } S \text{ permuted across components, draw } p) - \mathrm{MAE}_f \right], \tag{4.7}$$

the increase in component-weighted held-out error, in mSF2 units, when the information in that score is destroyed. Coefficient sign consistency across fits and the Spearman correlation between gene-set scores were also computed, so that shared signal between correlated sets is visible rather than hidden.

4.8 Pathway prioritisation

Gene sets were assigned to tiers by rules frozen before the run. *Tier C* requires that importance exceeded zero in at least half the fits and that the share of fits with a zero coefficient was at most 0.50. *Tier B* additionally requires sign consistency and a positive-importance share of at least 0.75 and significance against the permuted-outcome null; *tier A* requires both at least 0.90 and significance against both null models. **Both higher tiers also require a formally demonstrated prediction gain and at least 1,000 completed draws of each null.** Neither exists in this run, so tiers A and B were unreachable by construction, and the rules were not altered to produce a more impressive ranking. Scores correlated at $|\rho| \geq 0.80$ are flagged as sharing signal.

4.9 Cell-line triage

Each measured line was characterised by the absolute error of its repeat-averaged out-of-fold prediction under M1 and M2, the spread of its prediction across repeats, and the disagreement between the two models, each expressed as a cohort percentile (the share of lines with a strictly smaller value, plus one, divided by the number of lines). A line reaches *tier A* when its error is at or above the 95th percentile under both models, *tier B* at or above the 90th percentile under both, and *tier C* at or above the 90th percentile under either. Technical flags — stranded RNA profile, related lines in the cohort, a Cellosaurus problem flag on a relative, a small tissue group, unstable prediction and strong model disagreement — are recorded separately from the tier, so that a large residual caused by a data or identity problem is never presented as a biological discrepancy.

4.10 Curation of Moringa constituent evidence

Constituents and their biology were curated by literature search and reading only, into two validated evidence files. Only *primary* studies count, so a review citing an earlier study is never independent confirmation. Only studies of the **purified compound** can score; a study that dosed an extract is recorded with its score contribution disabled. Only *ionising* radiation qualifies as radiation evidence, which excludes the large UV photoprotection literature. Nanoparticle and metal-conjugate formulations are excluded because the carrier confounds the compound, and chemically distinct derivatives are never credited to the parent compound. The **extraction solvent of every report is recorded verbatim**, and each compound is classified as documented in a dichloromethane extract, documented in seed or leaf with another solvent, or indirect and unconfirmed. Every DOI was verified against Europe PMC or Crossref and every identifier against PubChem, with the verification route and reading depth stored per record.

4.11 Candidate scoring and tiers

Each compound receives four integer scores, fixed before any score was computed:

$$T = \underbrace{D_1}_{0\text{–}2} + \underbrace{D_2}_{0\text{–}3} + \underbrace{D_3}_{0\text{–}2} + \underbrace{D_4}_{0\text{–}1} \in \{0, \dots, 8\}, \tag{4.8}$$

where D_1 is the strength of verified *Moringa* constituent evidence (2 if at least one primary seed or leaf report has structural confirmation by isolation and NMR or by a reference standard; 1 if identification rests on a spectral-library match or is not stated; 0 if no primary seed or leaf report exists); D_2 the relevance to radiation-response biology (3 for ionising-radiation evidence from at least two independent research groups, 2 from one group, 1 if only a relevant process was modulated without radiation); D_3 independent source support (2 for at least three distinct research groups across both evidence files, 1 for two); and D_4 feasibility of follow-up (1 if the identifier is verified and at least one scored study reports a usable concentration or dose). Research groups are counted once across both evidence files, so repeated work from one laboratory is one source, and **chemical-class analogy earns no points**.

The tiers are A ($T \geq 6, D_1 \geq 1, D_2 \geq 2$), B ($T \geq 4, D_1 \geq 1, D_2 \geq 1$) and C ($T \geq 2, D_1 \geq 1$); compounds below these thresholds are not prioritised. Two safeguards override the score. A compound with no primary *Moringa* identification ($D_1 = 0$) is *not assessable* and never enters the shortlist. Evidence that a compound protects *tumour* cells from radiation, or is itself genotoxic, is a liability that is displayed and caps the tier at C. Extraction-specific evidence is reported beside the score and never folded into it.

Model-associated pathways, literature-supported processes and their overlap are kept apart. An overlap between the genes a published experiment measured and the members of a model-ranked gene set marks a hypothesis to be tested; it is not evidence that the compound modifies the gene set, the pathway or clonogenic SF2.

4.12 Reproducibility and quality assurance

The codebase uses Python 3.11 or later with a `src/` layout, type hints, small pure functions and YAML configuration with recorded seeds and no machine-specific paths. Every analytical output carries a run record with the git commit and a flag for uncommitted changes, configuration and file checksums, package versions and UTC timestamps, and analysis runs refuse to start from an uncommitted working tree. Result folders and figures are write-once, so a published number cannot be silently overwritten. Specialist reviews covering radiobiology, radiogenomic data curation, machine-learning methodology, statistics, software architecture, reproducibility, figure design and literature were recorded at each decision point in the project's decision log, together with any unresolved disagreement.

5 | Machine-learning experiments and results

5.1 The analysed cohort

The analysis cohort comprises **241 eligible records forming 237 independent biological components** across 20 tissues (Table 4; Figure 2a). Nine tissues have at least ten components and form their own cross-validation strata: Lung (46), CNS/Brain (23), Esophagus/Stomach (23), Bowel (21), Head and Neck (18), Breast (17), Pancreas (16), Bladder/Urinary Tract (11) and Ovary/Fallopian Tube (10). The remaining eleven tissues, from Skin (9) to Biliary Tract (1), are pooled. Seven components contain more than one record, and seven expression profiles are flagged as stranded RNA preparations.

Observed mSF2 spans almost the whole possible interval, and tissue medians differ visibly (Figure 2b), which is why a tissue-only model is the appropriate baseline against which to judge expression features. Two of the three cell lines of the motivating study are present: MCF-7 (observed mSF2 0.52) and DU145 (0.71). They are shown as context only. V79 is a Chinese hamster line and has no counterpart in this human cohort.

5.2 Model comparison

Table 5 and Figure 3 summarise the out-of-fold performance of the five prespecified models. Two results stand out.

First, **expression features reduce the error, and the reduction is small**. Relative to the tissue-only baseline, M2 lowers the mean absolute error by $\Delta\text{MAE} = 0.00383$ mSF2 units (Equation 4.6), or 2.85% of the baseline MAE, and it has the lower error in **nine of ten** cross-validation repeats; the exception, repeat 7, favours M1 by 0.0024 (Table S1). The proliferation-adjusted sensitivity comparison agrees in direction and size: M2P improves on M1P by 0.00387 and is better in all ten repeats.

Second, **no model predicts well**. Out-of-sample R^2 ranges from -0.018 to $+0.013$, so every model is approximately as accurate as predicting the cohort mean, and the tissue-only model is marginally *worse* than the global mean on both MAE and R^2.

Reconciliation of the reported difference. Three saved files report the primary difference, and they agree exactly once the estimand is named (Table 6). The first three entries are the same quantity computed in three ways. The record-weighted value differs by 1.5×10^{-5}, 0.4% of the estimate; it is not a discrepancy but a different weighting, which gives components with several records more influence. The protocol specifies the component-weighted estimate, and that value is used throughout. The relative figure of 2.85% is 0.0038317/0.1342282.

Table 4. Dataset composition by tissue of origin (OncotreeLineage). Components group related cell lines; tissues with fewer than ten components are pooled into one cross-validation stratum. Source: `results/tables/primary_v1/publication_v1/table_1_dataset_composition.csv`.

Tissue	Records	Components	Multi-record	Stranded RNA	Stratification
Lung	46	46	1	1	own stratum
CNS/Brain	23	23	1	2	own stratum
Esophagus/Stomach	23	23	0	0	own stratum
Bowel	23	21	2	1	own stratum
Head and Neck	18	18	0	0	own stratum
Breast	17	17	0	0	own stratum
Pancreas	16	16	0	1	own stratum
Bladder/Urinary Tract	11	11	0	0	own stratum
Ovary/Fallopian Tube	10	10	0	0	own stratum
Skin	9	9	0	0	pooled
Bone	9	8	1	1	pooled
Uterus	9	8	1	0	pooled
Pleura	6	6	0	0	pooled
Prostate	5	5	0	1	pooled
Kidney	3	3	0	0	pooled
Liver	3	3	0	0	pooled
Peripheral Nervous System	3	3	1	0	pooled
Soft Tissue	3	3	0	0	pooled
Thyroid	3	3	0	0	pooled
Biliary Tract	1	1	0	0	pooled
All	**241**	**237**	**7**	**7**	

Table 5. Out-of-fold performance of the five prespecified models. MAE, RMSE and changes in MAE are in mSF2 units; all values are component-weighted and averaged over ten repeats of grouped five-fold cross-validation. 'Change vs M1' and 'Change vs M0' are the reductions in MAE relative to those models (positive means a lower error); the paired ΔMAE is the primary estimand of Equation 4.6, computed against M1 for M2 and against M1P for M2P. Descriptive only: superiority was not formally assessed. Source: `results/tables/primary_v1/publication_v1/table_2_model_comparison.csv`.

Model	MAE	RMSE	R^2_{oos}	Slope	Int.	Change vs M1	%	Change vs M0	Paired ΔMAE	Repeats favouring
				calibration						
M0	0.133624	0.170755	−0.009497	−4.00	2.92	0.000605	0.45	0.000000	—	—
M1	0.134228	0.171494	−0.018275	−1.21	1.29	0.000000	0.00	−0.000605	—	—
M1P	0.134259	0.171304	−0.016026	−0.05	0.61	−0.000031	−0.02	−0.000636	—	—
M2	**0.130397**	**0.169115**	**0.009347**	0.65	0.21	0.003832	2.85	0.003227	**0.003832**	9/10
M2P	0.130388	0.168841	0.012822	0.63	0.22	0.003840	2.86	0.003235	0.003871	10/10

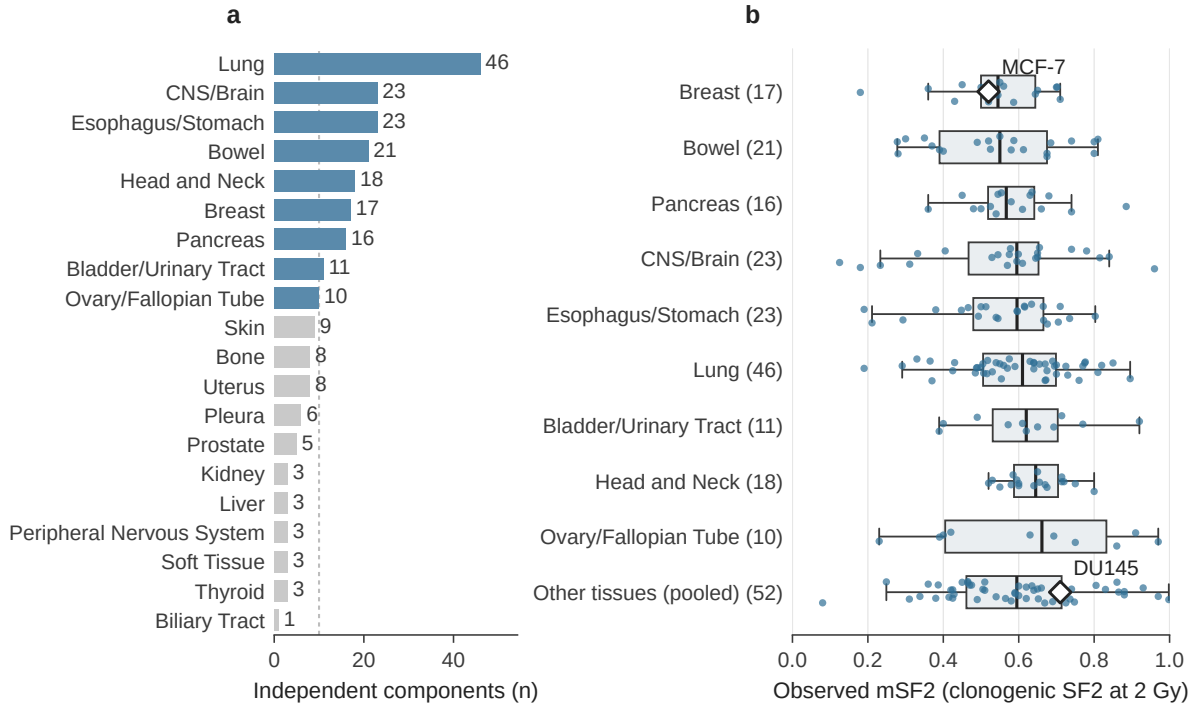


Figure 2. Composition and radiosensitivity of the analysed cohort (241 eligible records, 237 independent biological components). (a) Independent components per tissue (OncotreeLineage); the dashed line marks the ten-component threshold above which a tissue forms its own cross-validation stratum. (b) Distribution of observed CLARIFi mSF2, the literature-median clonogenic surviving fraction at 2 Gy, with one point per independent component so that related cell lines count once; tissues with fewer than ten components are pooled. Boxes show the median and interquartile range; whiskers extend to 1.5 times the interquartile range. Open diamonds mark MCF-7 and DU145, two of the three lines of the motivating *Moringa* study, shown as context only: no statement about *Moringa* extracts or radioprotection is made or implied.

Table 6. The primary difference as recorded in three saved output files.

Source	Value	Estimand
`comparisons.csv`, `delta_mae`	0.0038316558867076	Paired, component-weighted, mean over repeats (primary)
`model_metrics.csv`, MAE(M1) − MAE(M2)	0.0038316558867076	Difference of the two component-weighted MAEs
`per_repeat_delta_mae.csv`, mean of 10 repeats	0.0038316558867076	Mean of the ten per-repeat values
`comparisons.csv`, `record_weighted_delta_mae`	0.0038170756763974	Paired, record-weighted (each of 241 records weighted equally)

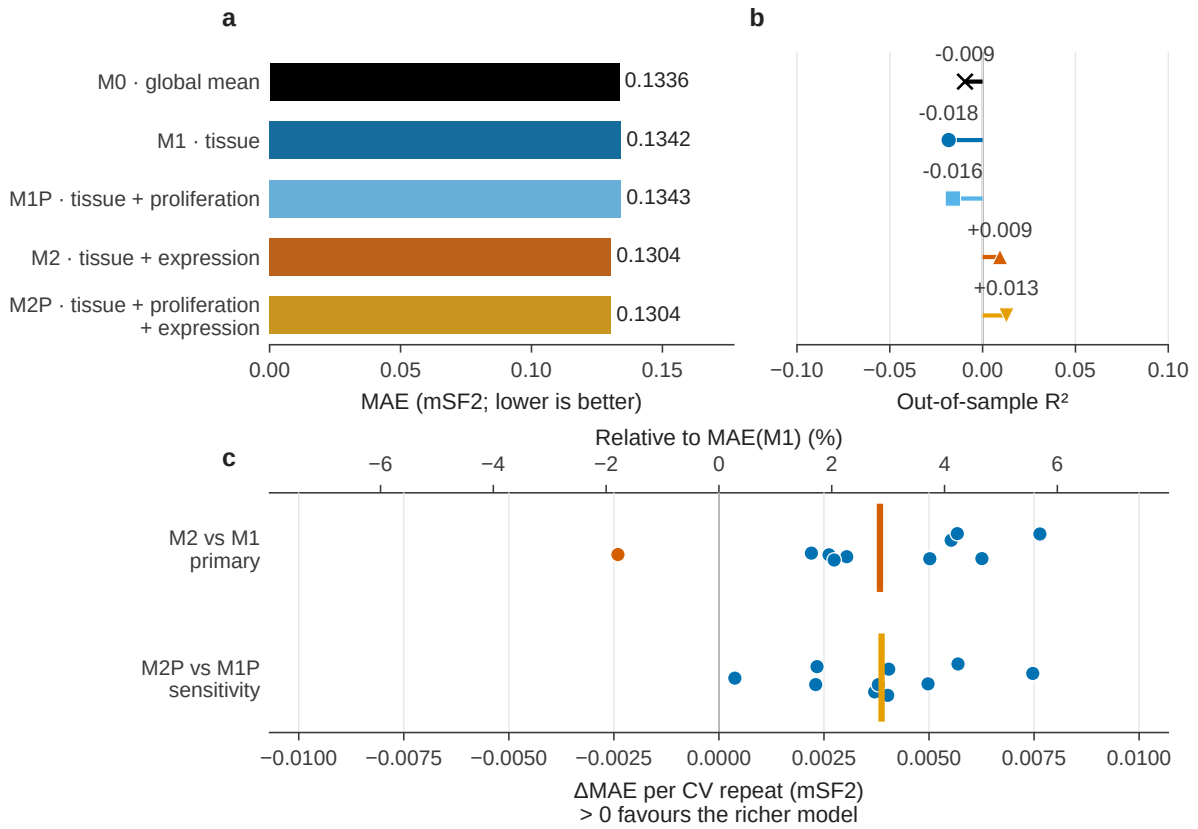


Figure 3. Out-of-fold predictive performance of the five prespecified models (M0 global mean, M1 tissue only, M1P tissue and proliferation, M2 tissue and 50 Hallmark expression scores, M2P tissue, proliferation and expression). (a) Component-weighted mean absolute error in mSF2 units, with bars starting at zero. (b) Out-of-sample R^2, where zero corresponds to predicting the cohort mean; every model lies close to zero. (c) Paired difference in MAE in each of the ten cross-validation repeats for the primary comparison (M2 against M1) and the proliferation-adjusted comparison (M2P against M1P); positive values favour the richer model, the thick bar is the mean over repeats, and the upper axis expresses the difference as a percentage of MAE(M1). Repeats re-split the same 237 components, so their spread shows split-to-split variability, not sampling uncertainty; no interval is shown because no interval method was validated for this design.

What the comparison establishes. The improvement is a descriptive estimate. Superiority was *not* formally assessed: the coverage simulation that would have validated an interval method was deferred (Section 10.2), so no interval or p-value is reported. No negative-control draws were run, so the gain cannot be placed against a permuted-outcome or random-gene-set reference. Repeated cross-validation is not external validation: the ten repeats re-split the same 237 components, so their spread measures split-to-split variability rather than sampling uncertainty, and says nothing about behaviour on an independent cohort.

5.3 Why out-of-sample R^2 stays near zero

Figure 4 shows the mechanism directly. Predictions from M1 and M2 occupy a narrow band close to the cohort mean, while the observations span almost the entire interval. The models are therefore less miscalibrated than uninformative: they shrink heavily towards the mean because the features support little more. The calibration slopes agree with this reading, moving from −4.00 for M0 and −1.21 for

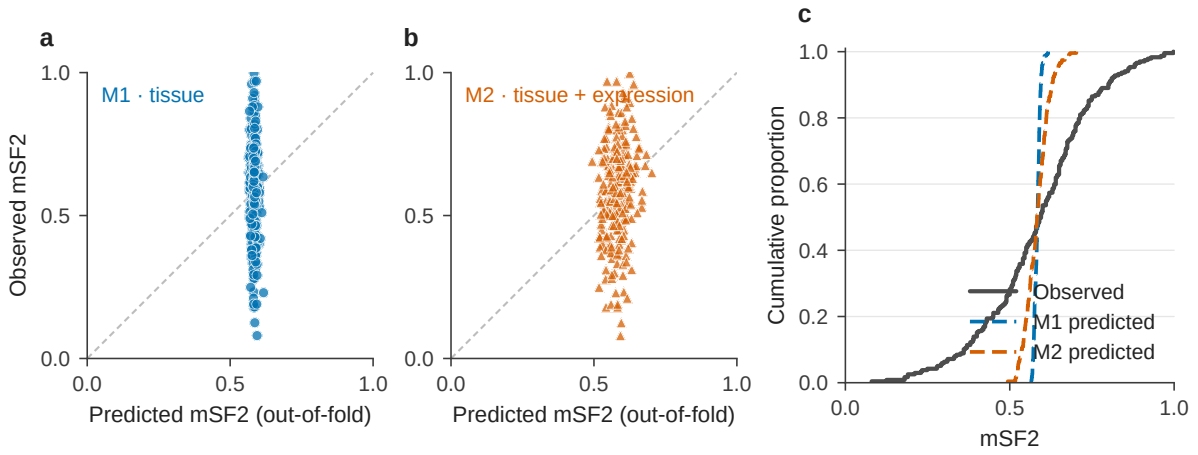


Figure 4. Why the out-of-sample R^2 stays near zero. (a, b) Observed against out-of-fold predicted mSF2 for the tissue-only model M1 and the tissue-plus-expression model M2. Each point is one independent biological component (records of a component averaged; each prediction already averaged over the ten repeats). The dashed line is the identity, and both axes span the full range of the endpoint. (c) Empirical cumulative distributions of the observed values and of the predictions of both models. The predictions occupy a narrow band near the cohort mean while the observations span almost the whole interval: the behaviour of a model with little usable signal rather than of a miscalibrated one. No trend line is fitted.

M1 to 0.65 for M2 and 0.63 for M2P (Table 5); the strongly negative slopes of M0 and M1 are an artefact of their predictions being nearly constant within a tissue (Section 5.5).

5.4 Tissue-level and repeat-level behaviour

Across the nine well-supported tissues, M2 achieves the lower error in **eight** (Table 7; Figure 5). The largest improvements are in Ovary/Fallopian Tube (0.0112), Breast (0.0084) and Head and Neck (0.0075). The exception is instructive: in **Pancreas, M2 is worse by 0.0110**, nearly three times the size of the pooled gain. A pooled improvement of this size is therefore not uniform across tissues, and it would be misleading to present it as a general property of the model. Estimates for tissues with fewer than ten components are shown in Table 7 for completeness but are unstable and are not interpreted.

Prediction stability across repeats (Figure S1) is higher for the tissue-only models, whose prediction is a shrunken tissue mean, than for the expression models, as expected.

5.5 A reporting correction

The first version of the within-tissue Spearman analysis reported strongly negative rank correlations for M1 in all nine larger tissues. Inspection of the code established the cause. Every fitted M1 predicts a single value per tissue, the shrunken training-fold tissue mean, so a held-out line's pooled out-of-fold prediction is effectively the mean of the *other* lines of its tissue, which falls as the line's own value rises. This is a leave-fold-out mean artefact, not a biological finding, and for M0 and M1 the statistic measures no within-tissue ranking at all. The corrected table (Table S2) reports the statistic as undefined wherever a model's prediction is constant within a tissue, and its defined values carry an explicit downward-bias caveat. The original pooled within-tissue correlations are not interpreted anywhere in this report; the first version is retained unchanged for audit.

Table 7. Component-weighted out-of-fold MAE by tissue (mSF2 units) and the difference ΔMAE = MAE(M1) − MAE(M2); positive values mean the expression model had the lower error. Tissues below the rule have fewer than ten components; their estimates are descriptive only and unstable. Source: `results/tables/primary_v1/publication_v1/table_3_tissue_performance.csv`.

Tissue	Components	MAE M0	MAE M1	MAE M2	ΔMAE	Direction
Lung	46	0.1251	0.1258	0.1249	0.0009	M2 lower
CNS/Brain	23	0.1558	0.1580	0.1538	0.0042	M2 lower
Esophagus/Stomach	23	0.1252	0.1279	0.1277	0.0002	M2 lower
Bowel	21	0.1512	0.1511	0.1471	0.0040	M2 lower
Head and Neck	18	0.0756	0.0720	0.0645	0.0075	M2 lower
Breast	17	0.1051	0.1050	0.0966	0.0084	M2 lower
Pancreas	16	0.0921	0.0934	0.1044	−0.0110	**M2 higher**
Bladder/Urinary Tract	11	0.1252	0.1255	0.1202	0.0053	M2 lower
Ovary/Fallopian Tube	10	0.2216	0.2272	0.2160	0.0112	M2 lower
Skin	9	0.1601	0.1630	0.1653	−0.0023	M2 higher
Bone	8	0.1287	0.1317	0.1290	0.0027	M2 lower
Uterus	8	0.1866	0.1791	0.1484	0.0308	M2 lower
Pleura	6	0.1855	0.1802	0.1788	0.0014	M2 lower
Prostate	5	0.1078	0.1102	0.0828	0.0275	M2 lower
Kidney	3	0.1627	0.1674	0.1666	0.0008	M2 lower
Liver	3	0.0430	0.0435	0.0696	−0.0261	M2 higher
Peripheral Nervous System	3	0.2041	0.2036	0.1985	0.0051	M2 lower
Soft Tissue	3	0.1117	0.1109	0.1255	−0.0146	M2 higher
Thyroid	3	0.1556	0.1576	0.1306	0.0270	M2 lower
Biliary Tract	1	0.4168	0.4146	0.3728	0.0418	M2 lower

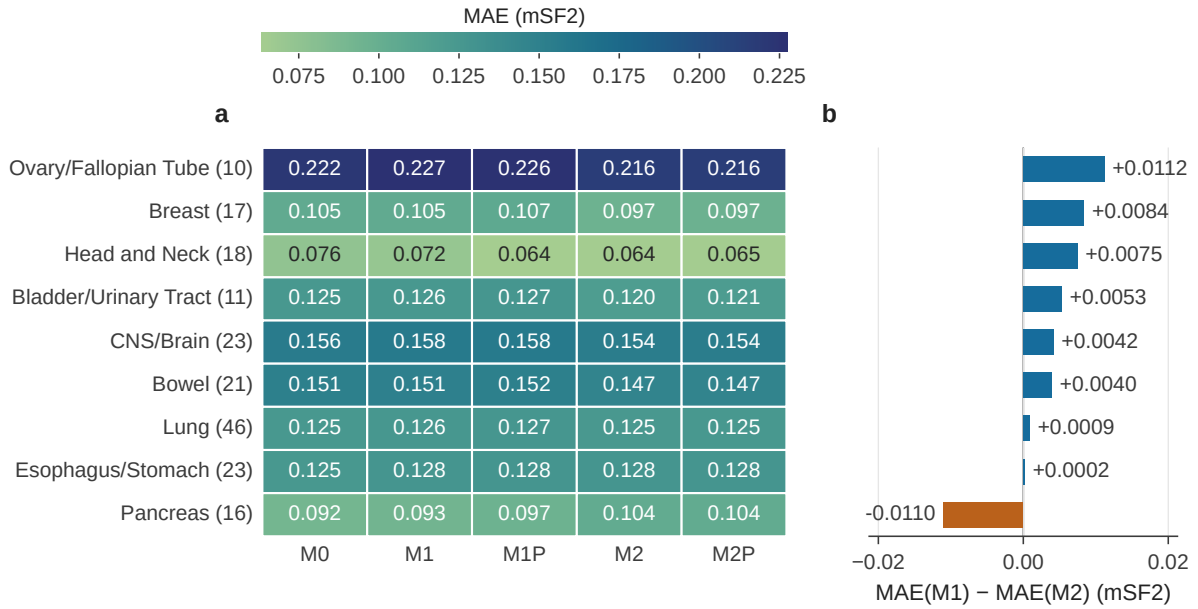


Figure 5. Performance by tissue of origin for the nine tissues with at least ten independent components; the number of components is given beside each tissue name. (a) Component-weighted out-of-fold MAE of each model, in mSF2 units; lighter cells are lower error. (b) The difference MAE(M1) − MAE(M2) for the same tissues in the same order; positive values (blue) mean the expression model had the lower error. Differences are small relative to the errors themselves and descriptive only: they come from one cross-validated run and no per-tissue inference was performed.

6 Robustness and tissue-generalisation analyses

The results of Chapter 5 come from the frozen primary benchmark. A separate and clearly labelled layer of analyses tests how durable its small advantage is, without altering it (Table 8). Three of the four planned analyses were executed; the negative controls were configured but not run.

6.1 A different model family does not do better

The random forest M3, fitted on the same splits with the same nested tuning, reached MAE 0.13148 and out-of-sample R^2 +0.00643, against 0.13040 and +0.00935 for M2 (Table 9). Its gain over the tissue-only baseline, ΔMAE = 0.00275, is **smaller** than the 0.00383 of M2. **M3 therefore performs worse than M2, and that is the result reported.** Within the same run, M0, M1 and M2 reproduced the primary run to the last decimal place, which also confirms that the frozen benchmark is reproducible.

6.2 Transfer to an unseen tissue is weak and conditional

Holding out each of the nine well-supported tissues in turn (185 components, 187 records), the expression-only model M2X reached MAE 0.12267 and out-of-sample R^2 **+0.0308**, against 0.12753 and −0.0084 for the global mean (ΔMAE = 0.00487 mSF2 units). This is the only analysis in the project in which out-of-sample R^2 is clearly above zero. It is, however, a different and easier comparison than the primary one — expression alone against a global mean, with tissue removed from both sides — so it is not comparable with the primary 0.00383 and does not revise it.

The pooled figure conceals an uneven picture (Table 10). The gain is largest when Ovary/Fallopian Tube (+0.0156) or Head and Neck (+0.0153) is held out, followed by Breast (+0.0119), Bowel (+0.0092), Bladder/Urinary Tract (+0.0079), Lung (+0.0049) and CNS/Brain (+0.0025). It is **negative for two tissues, Pancreas (**−0.0142**) and Esophagus/Stomach (**−0.0030**)**, where the expression-only model

Table 8. The exploratory and robustness layer: questions, runs and execution status.

Analysis	Question	Run	Status
Random forest M3	Does a different model family extract more signal from the same features and splits?	`exploratory_m3_v1`	Completed, 200/200 tasks
Leave-one-tissue-out	Does an expression-only model transfer to a tissue it has never seen?	`loto_v1`	Completed, 18/18 tasks
Outcome-scale sensitivity	Does the advantage survive the neglog and logit scales, or is it a raw-scale artefact?	`scales_v1`	Completed, 400/400 tasks
Negative controls (1,000 draws of each null)	Is the advantage larger than the pipeline produces when expression cannot carry signal? Are any gene sets more important than matched random sets?	`controls_v1`	Configured, about 10–20 h at six workers; **not executed**

Table 9. Exploratory model comparison including the random forest M3, on the raw mSF2 scale (MAE and RMSE in mSF2 units). M0, M1 and M2 reproduce Table 5. Source: `results/tables/exploratory_m3_v1/model_metrics.csv`; the paired differences quoted in the text are from `comparisons.csv` in the same folder.

Model	MAE	RMSE	R^2_{oos}	Calibration slope	Calibration intercept
M0	0.133624	0.170755	−0.009497	−4.00	2.92
M1	0.134228	0.171494	−0.018275	−1.21	1.29
M2	0.130397	0.169115	0.009347	0.65	0.21
M3	0.131480	0.169394	0.006434	0.55	0.26

Table 10. Leave-one-tissue-out performance of the expression-only model M2X by held-out tissue. ΔMAE is MAE(global mean) − MAE(M2X) in mSF2 units, so positive values favour expression; the Spearman correlation is that of M2X within the held-out tissue. The pooled row is from `comparisons.csv` and `model_metrics.csv` of the same run. Source: `results/tables/loto_v1/held_out_tissue.csv`.

Held-out tissue	Components	ΔMAE	Mean prediction shift	Spearman ρ
Bladder/Urinary Tract	11	0.0079	−0.0425	0.327
Bowel	21	0.0092	0.0096	0.231
Breast	17	0.0119	0.0290	0.324
CNS/Brain	23	0.0025	0.0735	0.359
Esophagus/Stomach	23	**−0.0030**	0.0273	−0.124
Head and Neck	18	0.0153	−0.0284	0.187
Lung	46	0.0049	−0.0221	0.178
Ovary/Fallopian Tube	10	0.0156	−0.0452	0.576
Pancreas	16	**−0.0142**	0.0115	−0.062
Pooled (nine tissues)	185	0.0049	—	—

predicted worse than the global mean. The Pancreas loss is nearly three times the pooled gain, and Pancreas is also the one well-supported tissue in which M2 was worse than M1 in the primary benchmark (Table 7). Transfer of expression information to an unseen tissue is therefore conditional on the tissue, and the tissue-level results, not only the pooled gain, belong in any statement about generalisation.

6.3 The advantage depends on the outcome scale

The most important negative finding of the robustness layer concerns the outcome scale. Repeating the M2-versus-M1 comparison with 20 repeats on two transformed scales gave ΔMAE = 0.00023 on the neglog scale — essentially nothing, with the out-of-sample R^2 of M2 falling to −0.0322 — and 0.00980 on the logit scale (Table 11). The direction favours M2 on all three scales. The magnitudes, however, are expressed in the units of each transformed scale, not in mSF2, so they cannot be compared directly with the raw-scale 0.00383; each must be read against the baseline error on its own scale, which is also given in the table. On the neglog scale the advantage is negligible; on the logit scale the out-of-sample R^2 of M2 is slightly positive (0.0045) while that of M1 is negative. The raw scale remains the protocol's prespecified primary scale, and these results are reported beside it rather than replacing it. Together they mean that the primary estimate should be read as **small, consistent in direction and sensitive to how the outcome is expressed**.

Table 11. Outcome-scale sensitivity of the primary comparison (M2 against M1). ΔMAE and MAE are in the units of each scale and are not comparable across scales. The raw row is the primary run (10 repeats); the transformed scales used 20 repeats on the same 237 components and 241 records. Descriptive only; no confirmatory method was adopted. Sources: `results/tables/scales_v1/comparisons.csv` and `model_metrics.csv`; raw scale from Table 5.

Scale	ΔMAE	Record-weighted	Repeats	MAE M1	R^2_{oos} M1	R^2_{oos} M2
raw (primary)	0.00383	0.00382	10	0.1342	−0.0183	0.0093
neglog	0.00023	0.00017	20	0.2596	−0.0164	−0.0322
logit	0.00980	0.00957	20	0.6398	−0.0142	0.0045

6.4 Rules governing the exploratory layer

Four rules keep this layer separate from the primary benchmark.

It cannot promote a new primary model. M3 was configured *after* the primary results were seen, so M2 remains the primary benchmark whatever M3 scores. In the event M3 scored worse, which is the expected outcome for a tree ensemble on 237 components and 50 coarse features and suggests that the ceiling is set by the data rather than by the model family.

Consistency with the frozen run is verified mechanically. A dedicated check confirms that an exploratory run covers the same 241 records and 237 components; that no biological component straddles a training and test fold or two inner folds; that the split table and its checksum are identical to those of the primary run; that the input checksums, base seed and fold counts match; that tuning is nested; that models shared with the primary run reproduce its predictions exactly; and that the primary run's own files still match their recorded checksums.

Negative controls would not by themselves create inference. With 1,000 draws, the gene-set p-values that are currently empty would become available, and the observed ΔMAE could be placed descriptively against a permuted-outcome null. Tiers A and B would stay closed, because that gate also requires a formally demonstrated prediction gain, which needs the interval method that the deferred coverage simulation was designed to validate. A permutation test would be a legitimate alternative route, but adopting it requires a protocol amendment recorded *before* the null distribution is inspected; written afterwards, it would be a post-hoc test chosen in sight of the result and would have to be reported as such.

The within-tissue correlation artefact is avoided. None of these analyses reports the pooled within-tissue rank correlation produced by the first version of that table; the corrected statistic is undefined wherever a model predicts a constant within a tissue (Section 5.5).

7 | Explainability and pathway prioritisation

7.1 Gene-set importance

All 50 fits of M2 completed and were analysed. Ranked by mean held-out permutation importance (Equation 4.7; Figure 6a; Table S3), the leading gene set is **HALLMARK_P53_PATHWAY**, with mean importance 0.00343 mSF2 units, importance above zero in 90% of fits, sign consistency 0.92 and a predominantly negative coefficient. It is followed by COMPLEMENT (0.00118), OXIDATIVE_PHOSPHORYLATION (0.00107), UV_RESPONSE_UP (0.00100), MYC_TARGETS_V2, KRAS_SIGNALING_UP and UV_RESPONSE_DN.

Three observations matter more than the ranking itself.

- **The magnitudes are tiny.** The largest mean importance, 0.0034 mSF2 units, is of the same order as the entire predictive gain, so the ranking describes a very small effect.
- **Stability is modest.** Across the 50 fits, the mean Kendall τ between importance rankings is 0.225, the mean rank-biased overlap 0.518 and the mean top-10 Jaccard index 0.391: the same gene sets do not reliably appear at the top of every fit.
- **Correlated sets share signal.** Several ranked sets are correlated with another set at $|\rho| \geq 0.80$ (Figure S2), including COMPLEMENT with APOPTOSIS and MYC_TARGETS_V2 with MYC_TARGETS_V1, so importance may be shared or exchanged between them rather than attributable to one.

Equally important is what the ranking does *not* contain. **DNA repair, G2M checkpoint and hypoxia, the pathways a radiobiologist would expect, were not prioritised**: G2M checkpoint ranked 13th with importance above zero in only 8% of fits, hypoxia 10th with 10%, and DNA repair 47th. Reporting this guards against presenting a familiar radiation-response narrative that the data do not support.

7.2 The pathway shortlist

Seven of the 50 gene sets reached **tier C** and 43 were not prioritised; **no set reached tier A or B**, because both require a formally demonstrated prediction gain and at least 1,000 negative-control draws, and the run has neither (Figure 6b). The gene-set tests record zero completed draws of both the permuted-outcome and the random-gene-set nulls, so attribution is not permitted and the stability gate cannot be evaluated.

The shortlist has a precise meaning. Its members are the gene-set scores whose permutation importance was positive in at least half of the cross-validation fits and which received a non-zero coefficient in at least half, within one model family on one dataset. It is not a gene-level finding, a validated biomarker, a causal mechanism, a radiation-response pathway or a molecular effect size, and it is not evidence that modulating any of these pathways would change radiosensitivity.

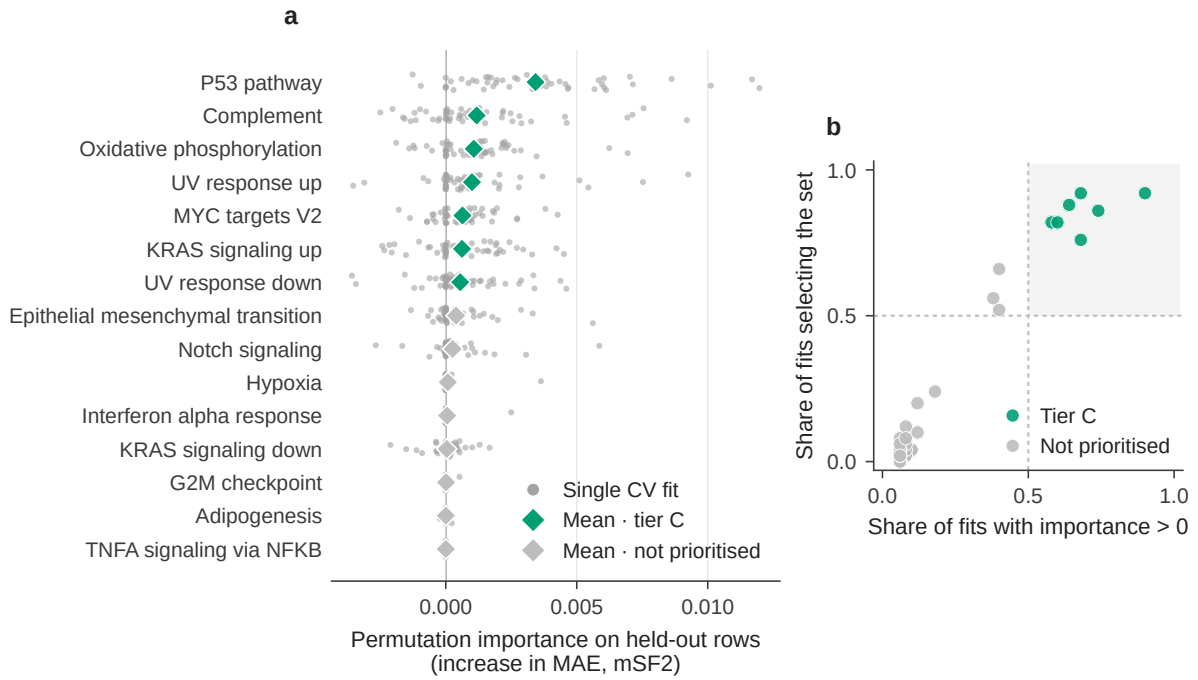


Figure 6. Exploratory importance of the Hallmark gene-set scores in the expression model M2. (a) The fifteen highest-ranked gene sets. Grey points are the grouped permutation importance in each of the fifty model fits (ten repeats by five folds), computed on held-out rows only as the increase in component-weighted MAE when a score is permuted across components; diamonds are the means over fits, coloured by the prespecified tier. The spread across fits is split-to-split variability, not a confidence interval. (b) The two tier-C criteria for all fifty gene sets: the share of fits with importance above zero and the share of fits in which the set received a non-zero coefficient; the shaded corner satisfies both thresholds (dashed lines at 0.50). Tiers A and B were unreachable by design. Shares are proportions of overlapping cross-validation fits, not probabilities. The rankings describe one fitted model family on one dataset; they are not validated biomarkers, causal mechanisms, radiation-response pathways or molecular effect sizes.

7.3 Cell lines the model reproduces least well

Of the 241 measured lines, 12 reached discrepancy tier A, nine tier B and 27 tier C; 193 were not prioritised (Figure 7; Table S4). The highest-ranked discrepancies are dominated by lines with extreme observed values — HT-144 (observed 0.08, predicted 0.59 under M2), KNS-42 (0.125 against 0.60), HEC-1-B (0.97 against 0.56) and D283 Med (0.18 against 0.58) — which is the shrinkage of Figure 4 seen one line at a time rather than a set of biologically unusual cell lines.

The triage separates possible explanations. Several tier A and B lines carry technical flags (small tissue group, related lines in the cohort, a Cellosaurus problem flag on a relative) and are routed to technical review first; the remainder are labelled as unexplained discrepancies requiring follow-up. Because the tier thresholds are cohort percentiles, a roughly fixed fraction of lines is always flagged: a flag means ‘relatively large in this cohort’, not ‘abnormal’.

Of the two context lines from the motivating study, **MCF-7 was not prioritised**: its held-out errors lie within the bulk of the cohort (observed 0.52 against 0.574 under M1 and 0.534 under M2). **DU145 reached tier C** with four technical flags — stranded RNA profile, small tissue group, unstable prediction across repeats and strong model disagreement — and was observed at 0.71 against predictions of 0.580 under M1 and 0.650 under M2. Both tiers were assigned by the same rule as for every other line. No

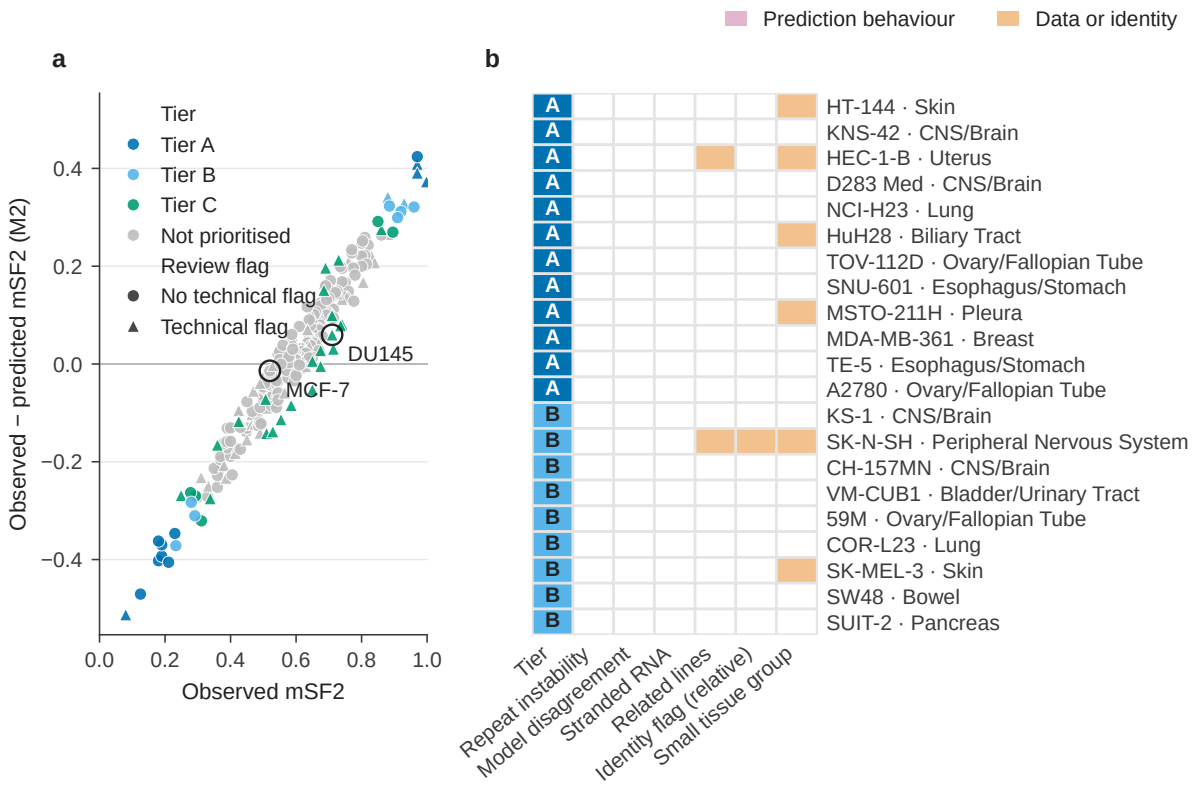


Figure 7. Cell lines whose measured radiosensitivity the model reproduces least well. (a) Residual (observed minus predicted mSF2 under M2, predictions averaged over repeats) against the observed value for every measured line, coloured by prespecified discrepancy tier; triangles mark lines with at least one technical flag. The strong dependence of the residual on the observed value is shrinkage: because the models predict close to the cohort mean, lines with extreme measured values necessarily have the largest residuals. Circled points are MCF-7 and DU145, labelled for their relevance to the motivating study; their tiers are not altered. (b) Flags recorded for the tier A and B lines, in rank order, separating flags about prediction behaviour (instability across repeats, disagreement between M1 and M2) from flags about the data or cell-line identity (stranded RNA profile, related lines in the cohort, a Cellosaurus problem flag on a relative, small tissue group). These are measured lines used in model development, not independent validation samples; a large residual is a signal to re-check identity, provenance and the original measurement, not evidence of unusual radiobiology.

individual prediction error is evidence of radioprotection, radiosensitisation or any causal mechanism, and lines selected for extreme residuals are expected to agree better on re-measurement even if nothing is wrong, through regression to the mean.

8 | Moringa compound evidence and candidate prioritisation

8.1 Verified constituents of Moringa oleifera seed and leaf

Curation produced a provenance-carrying inventory of *M. oleifera* seed and leaf constituents drawn from primary analytical studies, with the extraction solvent, identification method, DOI verification route and reading depth of each report recorded (Table S5). Three findings are substantive in themselves.

Only one constituent is confirmed in the solvent the original study used. **abdelsayed2021jfda** isolated **niazimicin**, a thiocarbamate glycoside, from the **seed methylene-chloride fraction**, confirmed its structure by HRMS, UV, IR, EI-MS and one- and two-dimensional NMR, and quantified it by validated UPLC-MS/MS in both the seed and the leaf CH_2Cl_2 fractions. The same study did *not* detect niazirin or glucomoringin there. The only other dichloromethane-specific identifications are GC-MS spectral-library matches in leaf DCM material, read at abstract level: 4-hydroxyphenylacetonitrile and methyl 4-hydroxyphenylacetate (**mohdfisall2021**), and palmitic acid and *cis*-vaccenic acid (**khandelwal2019**). The latter appeared in a journal indexed in neither PubMed nor Europe PMC, its full compound list could not be accessed, and electron-impact library matching cannot distinguish the Δ^9 and Δ^{11} C18:1 isomers without authentic standards.

The identification of several widely cited 'Moringa compounds' is weaker than their reputation. Free quercetin and kaempferol rest only on tentative LC-HRMS dereplication without authentic standards and may form by hydrolysis of the abundant glycosides during extraction; the forms that primary analytical work actually establishes in leaf are the **glycosides** isoquercitrin, astragalin and rutin. Under the prespecified rules the tentative aglycone reports still count as primary seed or leaf reports, so quercetin and kaempferol *are* assessable, but only in the weakest identification band ($D_1 = 1$). The pharmacokinetic caution applies in both directions: aglycone evidence may not be transferred to a glycoside, or the reverse, without explicit justification. Benzyl isothiocyanate is **not** a documented seed or leaf constituent: its precursor, benzyl glucosinolate, is reported in *roots*, while seed and leaf carry the rhamnosyloxy-benzyl glucosinolate instead. Gallic acid and caffeic acid yielded no primary report in *M. oleifera* seed or leaf and do not appear in the inventory.

Identifier hygiene was not a formality. Four public records returned by a name search are wrong or ambiguous for this purpose, and using them would have propagated an error: a mis-structured carbamate record ($C_{16}H_{23}NO_8$, with the ethoxy group on the sugar); a 'niaziminin A' record whose structure is not the thiocarbamate; a niazimicin record without stereochemistry; and glucomoringin, for which every available record is a potassium salt with a different stereo-InChIKey and none could be matched to the natural structure. The glucomoringin identifier remains **unresolved**.

8.2 The candidate shortlist

The evidence linking these constituents to radiation-response biology was curated separately, from primary studies of purified compounds only (Table S6), and the prespecified score of Equation 4.8 was then applied (Table 12; Figures 8 and 9). **Eighteen compounds entered the ranked shortlist and five reached tier A**; three further constituents scored below the tier-C threshold and were not ranked, and benzyl isothiocyanate was not assessable. The written justification of every point is given in Table S7.

Three features of the ranking matter more than the order itself.

The liability cap did real work. Chlorogenic acid scored 7 of 8, equal to four tier-A compounds, and is nevertheless held at tier C because one primary study reports that it *protects hepatocellular carcinoma cells* from radiation. A score alone would have promoted it; the prespecified cap prevented that.

The exclusion rule did real work. Benzyl isothiocyanate scored 6 on biology and sources and is still *not assessable*, because it has no primary report as a seed or leaf constituent. It cannot be ranked however strong its radiation evidence is.

The mismatch exposed by the curation is visible in the ranking. Niazimicin holds the only full identity score among compounds confirmed in a dichloromethane fraction ($D_1 = 2$) and scores *zero* on radiation biology, so it sits at tier C. Moringin, the signature *Moringa* isothiocyanate, reaches only tier B. Four of the five tier-A compounds hold $D_1 = 1$, a spectral-library or dereplication match, and none of the five has been found in a dichloromethane extract.

8.3 What the published biological evidence shows

The curated evidence establishes the following.

- **Moringin, the isothiocyanate most characteristic of *Moringa*, has no ionising-radiation study.** Exhaustive searching found no primary study combining moringin or glucomoringin with ionising radiation. Its evidence is confined to work without radiation: pro-apoptotic effects involving p53, CDKN1A, BAX and caspases in neuroblastoma cells (**cirmi2019**), G2/M arrest in prostate cancer cells (**abdkarim2023**), NRF2 activation in a motor-neuron model (**gugliandolo2024**), and NF-κB inhibition with a myeloma xenograft (**brunelli2010**). Because chemical-class analogy earns no points, moringin gains nothing from the benzyl isothiocyanate radiation literature, and that analogy is treated as an untested hypothesis.
- **The best-evidenced radiation candidates are not *Moringa*-specific.** β-Sitosterol has normal-tissue radioprotection evidence from two independent groups (**li2007**; **elbakary2026**), as does rutin (**patil2017**; **thabet2018**), although one research group contributes to both compounds' evidence; α-tocopherol has a rat crypt-microcolony clonogenic result (**felemovicius1995**) and a randomised controlled trial in radiation mucositis (**ferreira2004**). All three are documented in *Moringa* seed or leaf, but **none has been found in a dichloromethane extract**, which is why the extraction evidence is shown beside the score rather than within it.

Table 12. Candidate ranking of documented *M. oleifera* seed and leaf constituents under the prespecified four-dimension score (Equation 4.8; D_1 identity 0–2, D_2 radiation biology 0–3, D_3 independent sources 0–2, D_4 feasibility 0–1). Extraction evidence is shown beside the score and is not part of it. Rows follow the ranking file. Every tier is exploratory and hypothesis-generating. Source: `results/tables/primary_v1/compounds_v1/table_c3_candidate_ranking.csv`.

Rank	Compound	Part	Extraction	D_1	D_2	D_3	D_4	Total	Tier	Liability
1	α-Tocopherol	both	other solvent	2	3	2	1	8	**A**	—
2	Chlorogenic acid	leaf	other solvent	1	3	2	1	7	**C** (capped from A)	protects tumour cells (CGA-01)
3	Kaempferol	both	indirect	1	3	2	1	7	**A**	—
4	Quercetin	both	indirect	1	3	2	1	7	**A**	—
5	Rutin	leaf	other solvent	1	3	2	1	7	**A**	—
6	β-Sitosterol	seed	other solvent	1	3	2	1	7	**A**	—
—	Benzyl isothiocyanate	other	indirect	0	3	2	1	6	not assessable	—
7	Niazirin	both	other solvent	2	1	2	1	6	B	—
8	Oleic acid	seed	other solvent	1	2	1	1	5	B	—
9	Astragalin	leaf	other solvent	1	1	2	1	5	C (capped from B)	genotoxic (AST-01)
10	Moringin	both	other solvent	1	1	2	1	5	B	—
11	Phytol	leaf	other solvent	1	1	2	1	5	B	—
12	Glucomoringin	both	other solvent	2	0	2	0	4	C	—
13	Niazimicin	both	**DCM**	2	0	2	0	4	C	—
14	Acetylated moringin[a]	leaf	other solvent	2	0	1	0	3	C	—
15	Niaziminin	leaf	other solvent	2	0	1	0	3	C	—
16	*O*-Ethyl rhamnosyloxybenzyl carbamate[b]	seed	other solvent	2	0	1	0	3	C	—
17	Isoquercitrin	leaf	other solvent	1	0	1	0	2	C	—
18	Palmitic acid	both	**DCM**	1	0	1	0	2	C	—
—	4-Hydroxyphenylacetonitrile	both	**DCM**	1	0	0	0	1	not prioritised	—
—	Methyl 4-hydroxyphenylacetate	leaf	**DCM**	1	0	0	0	1	not prioritised	—
—	*cis*-Vaccenic acid	leaf	**DCM**	1	0	0	0	1	not prioritised	—

Note. Extraction: **DCM**, identified in a dichloromethane extract or fraction of seed or leaf; other solvent, identified in seed or leaf with another solvent; indirect, identification indirect or unconfirmed. For benzyl isothiocyanate the plant part is 'other' because its only *Moringa* report concerns its precursor in roots. [a] 4-[(4′-*O*-Acetyl-α-L-rhamnosyloxy)benzyl] isothiocyanate. [b] *O*-Ethyl-4-(α-L-rhamnosyloxy)benzyl carbamate.

- **One candidate carries an explicit liability.** Chlorogenic acid has normal-tissue protection evidence in bone marrow (**abraham1993**) and testis (**abedpour2022**), but **yin2022** report that it *protects hepatocellular carcinoma cells* from radiation through NRF2 and conclude that it could hinder radiotherapy. Under the prespecified rule this caps its tier at C. Astragalin is capped for a related reason: its only DNA-damage-relevant primary signal is *pro-genotoxic* (**saito2012**), and the one astragalin 'radiation' paper concerns UVB, which is not ionising radiation.
- **An informative negative was retained.** Membrane oleate enrichment did **not** alter the clonogenic radiosensitivity of human lymphoid cells (**george1983**). This direct clonogenic test argues against prioritising oleic acid despite its abundance in seed oil.

Reading Figure 8. Down the extraction column and across the four dimensions, the *Moringa*-specific compounds with the darkest identity cell — niazimicin, glucomoringin, niazirin, niaziminin, the acetylated isothiocyanate and the carbamate glycoside, all with a structurally confirmed presence in seed or leaf — are precisely those whose radiation-biology cell is empty or nearly so. The one compound with full scores for both identity and radiation biology is α-tocopherol, a ubiquitous plant metabolite that has not been found in a dichloromethane extract; the other four tier-A compounds carry only one identity point. Panel (b) shows the effect of the liability rule: chlorogenic acid's total of 7 equals that of four tier-A compounds, yet its bar is coloured as tier C. A reader comparing bar lengths alone would place it among the leaders; the prespecified cap prevents that, and the figure shows the discrepancy rather than hiding it.

Reading Figure 9. Attributed compound by compound, the biological evidence is thin. Only a minority of cells in panel (a) are filled, and several shortlisted compounds have no ionising-radiation evidence on their row: moringin, the most *Moringa*-characteristic candidate, appears only in the non-radiation colour, and niazimicin has no scored row at all because its one biological study dosed an extract fraction rather than the purified compound. Panel (b) is deliberately sparse. Only 25 compound-by-gene-set overlaps were recorded (Table S8), 23 for shortlisted compounds and two for benzyl isothiocyanate, which is not assessable and is not shown. They are small: 14 involve a single gene and 9 involve two, and the largest are moringin with the p53 pathway (three genes: BAX, CDKN1A, TP53) and quercetin with the p53 pathway (five genes: BAX, CDKN1A, DDIT3, TGFB1, TP53). These overlaps are the thinnest kind of link in the project, and they mark where a hypothesis could be formed, not where an effect has been shown.

8.4 What a tier means, and what it does not

A tier records **the state of the published evidence about a compound**, assembled under rules fixed before any score was computed (Table 13). It is not a measure of radioprotection. Tier A means 'best documented', not 'most likely to work'. Four of the five tier-A compounds are ubiquitous plant metabolites whose radiation evidence comes entirely from outside *Moringa* research, and the direction of that evidence differs between them: quercetin and kaempferol have *radiosensitisation* evidence in tumour cells as well as normal-tissue protection, whereas β-sitosterol and rutin have normal-tissue protection only. Radioprotection and radiosensitisation are opposite effects and are never merged (Table S6).

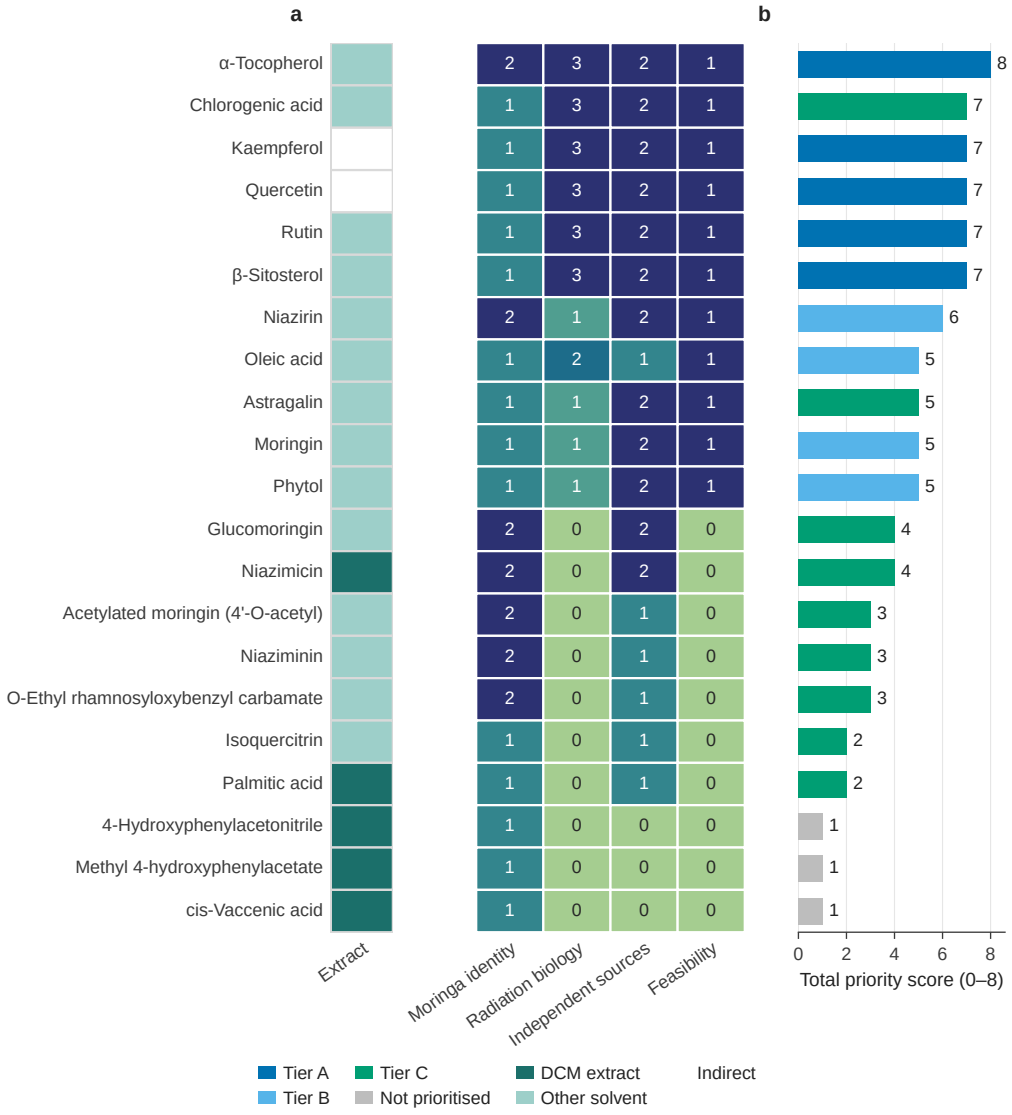


Figure 8. Evidence-based prioritisation of documented *Moringa oleifera* seed and leaf constituents for future experimental work. (a) Points awarded in each of the four prespecified dimensions (numbers are points; shading is the fraction of the dimension's maximum), beside the extraction-specific evidence: identified in a dichloromethane extract or fraction of seed or leaf (dark), in seed or leaf with another solvent (light), or only indirectly (white). (b) Total priority score out of eight, coloured by tier. The extraction column is kept outside the score because the original study used dichloromethane extracts whose composition was never measured. The scoring rules were fixed before any score was computed; every tier, including A, is hypothesis-generating, and no tier asserts radioprotection, radiosensitisation, potency or any effect of the original extracts. Two systematic names are shortened in the figure (see the note to Table 12).

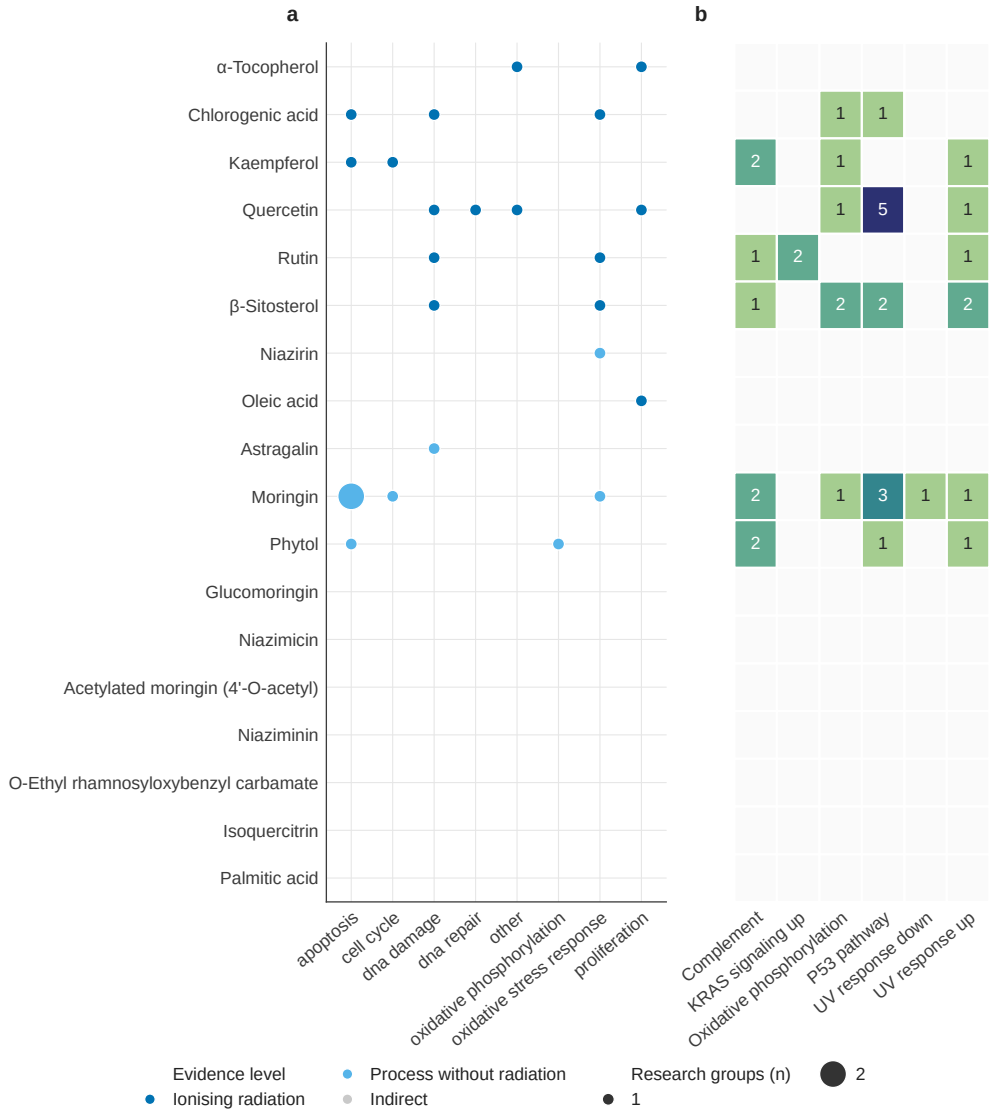


Figure 9. What the published evidence links each shortlisted compound to. (a) Radiation-response processes for which a primary study of the purified compound exists: symbol size is the number of independent research groups, and colour distinguishes studies using ionising radiation from studies of the same process without radiation. Compounds are in priority order. An empty row means that no qualifying primary study was located, which is a gap in the literature, not evidence of no effect. (b) Number of genes measured in a published experiment on the compound that are also members of a Hallmark gene set ranked by the exploratory explainability analysis (Table S8); blank cells mean no overlap was recorded. An overlap marks a point where a compound and a model-highlighted gene set touch the same genes. It is a hypothesis to be tested, not evidence that the compound modifies the gene set, the pathway or clonogenic SF2, and the gene-set ranking itself is capped at tier C. Reported directions of effect are given in Table S6 and are never merged.

Table 13. What a compound tier does and does not say.

A tier does say	A tier does not say
Primary studies of the purified compound exist, with the stated independence	That the compound protects or sensitises anything
The compound is documented in *Moringa* seed or leaf at the stated identification strength	That it was present in the extracts of the original study
Published work connects it to a radiation-response process	That it modifies clonogenic survival, the endpoint modelled here
A follow-up experiment can be specified from the record	That such an experiment would succeed

8.5 Realistic first-stage experimental priorities

The ranking answers the question 'what is best documented?'. A first experiment has to answer a narrower question, so the two orderings differ — without any score being changed. Three questions can be separated.

Is any plausibly active compound present in the extracts that were tested? This is the project's largest gap and the cheapest to close. The priority candidates are **niazimicin** (tier C, the only compound with a confirmed structural identification in both the seed and the leaf dichloromethane fractions) and **moringin** (tier B, the signature *Moringa* isothiocyanate). Neither has any ionising-radiation study, so neither can be offered as a protection candidate; they are priorities for **analytical characterisation of the original extracts**, followed by a first radiation screen if they are found.

Does a compound with existing radiation evidence act on the endpoint modelled here? **α-Tocopherol** is the strongest first-stage candidate: it is the only tier-A compound with a full identity score, and the only compound in the evidence base whose normal-tissue protection evidence includes a *clonogenic* endpoint (rat intestinal crypt survival), alongside a randomised trial in radiation mucositis. **β-Sitosterol** follows, with normal-tissue protection reported by two independent groups. Both must be tested in matched normal and tumour cells, because neither has a tumour arm.

Which candidates should not lead? **Chlorogenic acid** should not be a first-stage protection candidate despite its score: the tumour-protection finding makes it a cautionary compound in a radiotherapy context. **Astragalin** is capped for a related reason, its only DNA-damage-relevant evidence being pro-genotoxic. **Oleic acid** has a direct clonogenic test showing *no* effect on radiosensitivity, so it is informative but not a candidate. Quercetin, kaempferol and rutin are scientifically interesting but second-stage: their *Moringa* identity rests on tentative dereplication, and the aglycone–glycoside pharmacokinetic question would have to be settled first.

> **A defensible first-stage programme.** Characterise the original extracts (niazimicin, moringin), and test α-tocopherol and β-sitosterol on clonogenic survival in matched normal and tumour cells over a dose range. This uses the tiers as summaries of evidence, not as predictions, and leaves every score exactly as computed.

The scientific shape of this module's result is therefore a **mismatch**, and it is its most useful output: the compound with the best claim to being present in the tested extracts has no radiation biology, and the compounds with radiation biology have no confirmed presence in those extracts. Neither gap can be closed by further literature work; both can be closed by the experiments proposed in Chapter 12.

9 | Discussion

9.1 What the modest improvement means

Expression features reduced the error of a tissue-only baseline by 2.85%, consistently in direction (nine of ten repeats) but small in size and not uniform across tissues, with Pancreas moving in the opposite direction by nearly three times the pooled gain. The defensible interpretation is that **baseline Hallmark pathway activity carries a small amount of information about intrinsic radiosensitivity beyond tissue of origin — on this endpoint, in this cohort and under this model family**. This is not a demonstration that expression predicts radiosensitivity usefully, and it is not a validated effect: no interval, no p-value, no negative control and no external cohort stand behind it.

The robustness analyses sharpen this reading rather than soften it. The advantage survives a change of model family only in diminished form, since the random forest gains less (0.00275 against 0.00383). On transformed outcome scales it keeps its direction but nearly disappears on the neglog scale; a result whose size depends so strongly on the choice of transformation is small enough to be shaped by it. The one encouraging result, the leave-one-tissue-out R^2 of +0.031 for expression alone, comes from an easier comparison on a different design, and it reverses for two of the nine well-supported tissues (Pancreas −0.0142, Esophagus/Stomach −0.0030). Pancreas is unfavourable in both the primary and the leave-one-tissue-out analyses, so the tissue-level picture, and not only the pooled gain, must be part of any statement about generalisation.

Three further cautions apply. The comparison is nested (M2 contains M1 as an offset), so the errors of the two models are strongly correlated and the paired difference, while the right statistic, is a narrow one. Repeated cross-validation re-splits one fixed set of components and cannot estimate generalisation to new cohorts. And the baseline itself is weak: M1 is marginally worse than the global mean on MAE, which indicates that tissue of origin, as encoded here, contributes little once shrinkage is taken into account.

9.2 Why out-of-sample R^2 remains low

Figure 4 shows predictions compressed near the cohort mean against observations that span the whole interval. Several causes act together and cannot be separated with these data. The **label** is a literature median that aggregates laboratories, dose rates and protocols, and the field's own reporting audit found a third of papers omitting replicates or dose rate (**oike2020**); CLARIFi supplies one median per line, so per-line assay variability cannot even be estimated. The **sample** is small (237 independent components) and tissue-imbalanced. The **features** are 50 coarse pathway summaries of baseline expression, which may simply not encode the determinants of radiation response; intrinsic radiosensitivity may depend on dynamic, post-irradiation responses that a baseline profile cannot capture. And the **design is strict**: grouping by relatedness and confining all preprocessing to training folds remove the two most common sources of inflated performance.

This near-zero R^2 closely reproduces RadSigBench, which reported mean out-of-sample R^2 of 0.01 and 0.00 for seven published signatures and found them no better than random gene sets (**oconnor2022**). **Agreement with the strongest prior negative result is a substantive finding, not a failure**: it indicates that the pipeline measures what it claims to measure, and it is evidence against the more optimistic claims of the signature literature.

9.3 The value and uncertainty of expression-derived pathways

The pathway ranking is best read as a description of a weak fitted model. Its magnitudes are of the same order as the whole predictive gain, its rank stability across fits is modest (mean Kendall τ 0.225, mean top-10 Jaccard 0.391), and correlated gene sets share or exchange importance. The appearance of HALLMARK_P53_PATHWAY at the top is biologically *plausible*, since p53 governs the damage response, cell-cycle arrest and apoptosis, and its predominantly negative coefficient is consistent with higher p53-pathway activity accompanying lower survival. Plausibility is, however, precisely the trap: without a demonstrated prediction gain and without negative controls, a plausible ranking cannot be distinguished from a coincidence, and the absence of DNA repair, G2M checkpoint and hypoxia from the shortlist shows that the ranking is not simply recovering textbook radiobiology. The ranking is therefore reported as capped, exploratory and hypothesis-generating, and no mechanistic conclusion is drawn from it.

9.4 How the original Moringa findings motivate further work

The original study's lasting contribution is a *question* rather than a measurement this project could use: the post-irradiation proliferative response to a plant extract differed between cell types, most clearly for the leaf extract (PEF 1.26 in V79 against 0.90 and 0.99 in the two carcinoma lines). The computational work formalises the general version of that question on a valid endpoint and with proper grouping, and finds the molecular signal weak. This does not contradict the original observation, because the endpoints differ: a 72-hour biomass ratio measured with extract present is not intrinsic clonogenic survival. The original observation also remains confounded, since the only apparently normal line is the only non-human one. What the project adds is a defensible framework in which the question can be asked properly, together with an explicit statement of the experiments needed to answer it.

9.5 What compound prioritisation adds, and why the candidates remain hypotheses

The compound module turns an open-ended phytochemical literature into an auditable evidence trail: compound → documented presence in seed or leaf with a named solvent → primary experimental association with a radiation-response process → the genes actually measured. Its practical value lies as much in what it *excludes* as in what it ranks. It removes benzyl isothiocyanate (a root, not a seed or leaf, constituent), gallic and caffeic acid (no primary *Moringa* report), a UVB literature that is not radiation evidence, carrier-confounded nanoparticle studies and a retracted paper; it places the free flavonol aglycones in the weakest identification band rather than excluding them, as their tentative dereplication warrants; and it flags the tumour-protection liability of chlorogenic acid, which a naive 'antioxidants are radioprotectors' reading would have turned into a selling point.

Table 14. Results supported by the saved outputs, separated from plausible explanations that are not established.

Supported by the outputs	Plausible, not established
ΔMAE = 0.00383 (2.85%); M2 better in nine of ten repeats	That expression features would improve prediction in an independent cohort
Out-of-sample R^2 near zero for all five models	That richer features or more data would raise it materially
M2 lower error in eight of nine well-supported tissues, worse in Pancreas	That tissue-specific differences reflect tissue-specific biology
Expression-only transfer to unseen tissues: $R^2 = +0.031$, negative for Pancreas and Esophagus/Stomach	That expression generalises to tissues absent from training
Advantage keeps its direction on three outcome scales but nearly vanishes on one	That the raw-scale magnitude is the 'true' size of the effect
Seven gene sets meet the tier-C criteria; the p53 pathway ranks first	That the p53 pathway influences radiosensitivity in these lines
Importance rank stability is modest across fits	That a larger sample would stabilise the ranking
Niazimicin is confirmed in seed and leaf DCM fractions	That it was present in the original study's extracts
Moringin has no published ionising-radiation study	That it would behave like benzyl isothiocyanate under radiation
Chlorogenic acid protected hepatocellular carcinoma cells from radiation in one study	That it would protect human tumours during radiotherapy

The candidates remain hypotheses for specific reasons. No compound has been shown to be present in the extracts that were tested, because those extracts were never analysed. Moringin, the most *Moringa*-specific candidate, has no radiation study at all. The compounds that do have radiation evidence were mostly tested at concentrations or doses whose relevance to an extract is unknown, several at levels well above plausible dietary exposure, and many abstracts omit the radiation dose or the concentration. An antioxidant is not thereby a radioprotector; a compound that reduces tumour-cell viability does not thereby protect normal tissue; and a high priority score reflects the state of the published evidence, not a predicted experimental outcome.

The link to the machine-learning results is deliberately kept weak, because the evidence for it is weak. With the gene-set ranking capped at tier C, the compound prioritisation rests on external biological evidence, and any overlap between a compound's measured genes and a ranked gene set is reported as a hypothesis. **Nothing in this report claims that a prioritised pathway and a prioritised compound together imply an effect on clonogenic SF2.**

9.6 Separating supported results from plausible explanations

Table 14 separates what the saved outputs support from explanations that are plausible but not established. The left-hand column is the evidential content of this report; the right-hand column is the agenda for future work.

10 | Implementation, challenges and deliverables

10.1 Challenges and how they were addressed

An incomplete source manuscript. Printed page 3 and pages 7–12 of the original manuscript are missing, removing the extraction and culture methods, the end of the discussion, the conclusions and the references; authors, venue and DOI are absent, and no published version was found. In the manuscript's own figures the 0 Gy and 2 Gy series cannot be distinguished in the greyscale scan. The response was a labelled page-by-page extraction that marks every absent item as not stated, with no reconstruction and no use of point values read from the figures.

Endpoint confusion as a standing risk. The central scientific hazard of the project is to treat proliferation, viability and clonogenic survival as interchangeable. Endpoints are therefore separated in the data model, the documentation and the figure rules; a dedicated comparison of PEF with SF2 is recorded; and a project rule forbids placing different endpoints on one axis.

Dataset harmonisation and licensing. Four resources with different identifier namespaces, release cycles and licences had to be joined, and the outcome dataset is licensed for non-commercial use only. Each source has a manifest with checksums and access dates, the loaders verify checksums, and licences are recorded explicitly. The identity of the Hallmark gene-set file with the official release remains to be confirmed.

Cell-line identity and relatedness. Names are ambiguous, and relatedness is the most dangerous leakage route in this design. Matching used stable identifiers only; Cellosaurus hierarchy and origin links merged 241 records into 237 components; problem flags on relatives were recorded and surfaced in the triage; and the identifier, component and split code was audited for leakage.

A small sample against many features. With 237 independent components only modest model complexity is supportable. Simple scikit-learn baselines were used before anything more complex, within a prespecified ladder of five models with nested tuning, and the random forest was confined to an explicitly post-hoc exploratory configuration.

Computational cost. The validation plan required a coverage simulation to choose an interval method before any inference. The first pilot took about 24–25 minutes per replicate on one core, which made 9×500 replicates impractical within the available time. A static review found redundant work in the inner tuning loop: the gene-set scorer, the score scaler and the validation scores were being refitted for each of the 36 grid points in each of five inner folds, although they depend only on the fold rows, and the baseline ridge was refitted per grid point although it depends only on its own

Table 15. Analyses implemented and tested but not executed.

Analysis	Command	Status
Negative controls, 1,000 draws of each null	`scripts/run_model.py` with `configs/experiment_controls.yaml`	Configuration ready, cost measured at about 10–20 h; **not executed**. Tiers A and B stay closed and no gene-set p-value exists until it is run.
Coverage simulation (interval method)	`scripts/run_interval_simulation.py`	Infrastructure ready; **deferred** by the recorded protocol deviation.

penalty. A caching rewrite was designed to be bit-identical (same operations in the same order), with the original loop kept as the reference implementation for every model to which the cache does not apply. Bounded multiprocessing was added (spawn context, atomic write-once checkpoints written by the parent process only, one BLAS thread per worker), sized to the investigator's machine (six physical cores, twelve threads, about 31 GiB of memory). Even so, the full simulation could not be completed before the deadline.

10.2 The protocol deviation

The deviation was written into the decision log **before** the primary run produced a single result, which is what makes it a deviation rather than a rationalisation. Its terms are: the inference method is set to 'not assessable'; no interval or p-value is written; the number of repeats is fixed at ten; the negative controls are left at zero draws; and, as a consequence, gene-set attribution can never be permitted and the pathway tiers stay capped. The deviation is reflected in the results, the figures and the limitations, and the simulation infrastructure remains in the repository ready to run (Table 15).

A presentation error found and corrected. The first version of the within-tissue Spearman statistic was strongly negative for the tissue-only models. Rather than report a striking number, the code was inspected, the leave-fold-out mean artefact identified, and a corrected table written that marks the statistic undefined where a model's prediction is constant within a tissue (Section 5.5; Table S2). The first version was kept unchanged for audit and is not interpreted — a clear example of the write-once discipline doing its job.

Uncertainty in the correspondence between compounds and extracts. The original extracts were never chemically characterised, and the published dichloromethane-extract literature is thin, partly inaccessible and partly based on library matches. The response was a three-level extraction category per compound, kept outside the score; verbatim solvent wording stored for every report; the exclusion of speculation as evidence (one paper's 'possibly yes' for chlorogenic acid in a leaf CH_2Cl_2 extract); and a note of a citation error in a 2026 paper, in which a statement about an Indian DCM extract cites an unrelated soil-chemistry reference.

Abstract-level evidence extraction. Many full texts were paywalled or could not be retrieved, so radiation doses, concentrations and even cell-line identities are missing from several records. Every such gap is recorded as 'not stated' together with the reading depth, and the gaps are listed as items to close before publication rather than filled by inference.

10.3 Deliverables

The project delivered the following outputs.

1. A labelled extraction of the original manuscript, with every absent item marked and checked against the page images.
2. A provenance-tracked analysis dataset of 241 records and 237 independent components from 20 tissues, with checksum-verified inputs and a recorded quality-control pass.
3. The primary benchmark: 250/250 model-fold tasks for five prespecified models in ten repeats of grouped, nested five-fold cross-validation, with saved out-of-fold predictions, metrics, per-repeat differences and a run record.
4. The explainability analysis: held-out permutation importance for all 50 fits of M2, coefficient sign consistency, gene-set score correlations and rank-stability statistics.
5. Pathway and cell-line triage under enforced prespecified gates: seven tier-C gene sets and no tier A or B; 12, 9 and 27 cell lines in tiers A, B and C.
6. Publication tables derived only from saved outputs, including the corrected within-tissue ranking analysis with the superseded version retained for audit.
7. A curated, fully referenced compound evidence base (constituent inventory and primary radiation-biology evidence), with every DOI verified against Europe PMC or Crossref and every identifier against PubChem.
8. The prespecified compound-prioritisation protocol, its tested implementation and the resulting ranking: 18 compounds ranked, 5 in tier A, with the liability cap and the not-assessable rule both binding.
9. Three completed robustness analyses — the M3 random forest (200/200 tasks), leave-one-tissue-out (18/18) and outcome-scale sensitivity (400/400) — each reported as it came out, including where it weakened the primary reading, and M3 explainability kept separate from the primary analysis.
10. Eleven report figures as 600 dpi PNG and vector PDF with provenance records, and this report.

All model fitting, prediction, explainability and analysis were executed by the investigator; the automated assistant prepared code, configuration, documentation and this report, and ran no scientific computation.

11 | Limitations

1. **Predictive performance is weak.** Out-of-sample R^2 lies between −0.018 and +0.013 for every model. No model in this study predicts clonogenic radiosensitivity usefully, and the pipeline should not be presented as a working predictor.
2. **The improvement is small and was not formally tested.** ΔMAE = 0.00383 mSF2 units (2.85%). No interval, p-value or formal superiority assessment exists, because the coverage simulation that would have validated an interval method was deferred.
3. **No negative controls were run.** Zero permuted-outcome and zero random-gene-set draws were completed, so the gain cannot be compared with a null reference, gene-set attribution is not permitted and the stability gate cannot be evaluated.
4. **There is no external validation.** The ten repeats re-split the same 237 components; repeated cross-validation measures split-to-split variability, not generalisation.
5. **The improvement is not uniform, and transfer to unseen tissues is conditional.** In Pancreas, M2 is worse than M1 by 0.0110, nearly three times the pooled gain; in the leave-one-tissue-out analysis the expression-only model lost to the global mean when Pancreas (−0.0142) or Esophagus/Stomach (−0.0030) was held out.
6. **The size of the advantage depends on the outcome scale**, and magnitudes on the neglog and logit scales are in transformed units that cannot be compared directly with the raw-scale estimate.
7. **One outcome source with aggregated labels.** CLARIFi mSF2 is a literature median across laboratories, dose rates and protocols, with one value per line, so per-line assay variability cannot be estimated; the dataset is also licensed for non-commercial use only.
8. **A small, imbalanced sample.** There are 237 components; eleven of the 20 tissues have fewer than ten, and one has a single component.
9. **Pathway rankings are exploratory and only moderately stable.** Mean Kendall τ is 0.225 and mean top-10 Jaccard 0.391; importance magnitudes are the size of the whole predictive gain; correlated sets share importance; no tier exceeds C.
10. **Cell-line discrepancies are dominated by shrinkage.** Large residuals occur mostly at extreme observed values and are review signals, not biological findings; measured lines are not independent validation samples.
11. **No compound has been shown to be present in the original extracts.** The extracts were never analysed; most compounds are documented with other solvents; and several DCM identifications are spectral-library matches from inaccessible full texts, one in a journal indexed in neither PubMed nor Europe PMC.
12. **The most *Moringa*-specific candidate has no radiation evidence.** No primary study has combined purified moringin or glucomoringin with ionising radiation.
13. **Compound evidence is largely abstract-level.** Radiation doses, concentrations and some cell-line identities are missing from several records and must be recovered from the full texts; the glucomoringin identifier remains ambiguous; the erratum to **abedpour2022** has not been read;

and because the journal that published **yin2022** had a large contemporaneous retraction wave, that article's status must be re-checked (no retraction notice was found on 22 October 2026).

14. **Group independence is lower than paper counts suggest.** Shared senior authors and a shared compound supplier link several studies; the scoring counts research groups once, but the effective number of independent sources remains small.
15. **The within-tissue ranking statistic is unusable for the simpler models.** It is undefined where predictions are constant within a tissue and biased downward elsewhere.
16. **No causal or mechanistic claim is supported.** Association, explanation of a fitted model, causation and experimentally demonstrated protection are four different things, and only the first two are in evidence here.

12 | Recommendations for experimental validation

The following priorities are proposals; none has been carried out. They are ordered so that each removes an uncertainty on which the next depends.

Priority 1 — close the chemistry gap. Characterise the *original* dichloromethane seed and leaf extracts by LC-MS/MS and GC-MS against authentic standards, using the published validated UPLC-MS/MS method for niazimicin (**abdelsayed2021jfda**) as a positive control. Until this is done, no compound-level interpretation of the original result is possible. Yields and concentrations should be reported as measured, never inferred.

Priority 2 — resolve identities. Confirm the structures and public identifiers of the shortlisted compounds, settle the glucomoringin stereochemistry, obtain the paywalled full texts and recover the missing radiation doses and concentrations. Re-verify the retraction status of the flagged journal and read the outstanding erratum.

Priority 3 — complete the deferred statistical validation. Run the coverage simulation to choose and justify an interval method, re-assess the primary comparison with the adopted method, and run at least 1,000 draws of each negative control (permuted outcome and random gene sets). Only then can superiority be assessed, gene-set attribution be permitted or any pathway exceed tier C. The infrastructure for all of this is in place.

Priority 4 — test purified compounds against whole extracts on the right endpoint. Use multi-dose **clonogenic survival** assays (not 72-hour biomass) with and without compound or extract, yielding linear-quadratic parameters,

$$\mathrm{SF}(D) = \exp(-\alpha D - \beta D^2), \tag{12.1}$$

where α and β here denote the linear and quadratic radiobiological coefficients, and a dose-modifying factor for each treatment. Include vehicle-only and cell-free interference controls, which the original study did not describe, and independent biological repeats on separate days and passages, since the original replicates were technical wells and dose was confounded with plate.

Priority 5 — make the normal-versus-tumour comparison interpretable. Use species-matched and tissue-matched normal and tumour pairs assayed identically, so that species, tissue and malignancy are not confounded as they were when one hamster fibroblast line was compared with two human carcinoma lines. A selectivity claim requires treated normal *and* treated tumour measurements on the same endpoint. The best published model of how such a claim should be worded is the quercetin

study of **horton2013**, which reports normal-tissue anti-fibrotic protection with *no* effect on fibroblast or murine tumour radioresponse.

Priority 6 — test the prioritised pathways directly, if Priority 3 succeeds. Perturb the implicated processes (for example p53-pathway activity) in a small panel of cell lines spanning the predicted range, and measure clonogenic survival. This is hypothesis testing of a model-derived conjecture, not confirmation.

Priority 7 — strengthen the benchmark. Seek a second, independent radiosensitivity source for cross-assay evaluation, reported as heterogeneity rather than pooled. Consider prospective prediction on unassayed lines under a panel-selection rule written into the protocol *before* predictions are viewed, reported as a designed validation panel rather than an unbiased external cohort.

13 | Conclusions

This project set out to connect an unpublished *Moringa oleifera* irradiation study to a defensible computational investigation of cellular radiosensitivity, and to use the result to identify compounds worth testing.

What was established. A reproducible, provenance-tracked pipeline links public clonogenic radiosensitivity data to baseline expression for 237 independent biological components, with cell-line relatedness handled correctly and all preprocessing confined to training folds. Within that design, pathway-level expression features reduce the error of a tissue-only baseline by 0.00383 mSF2 units (2.85%), consistently in direction across nine of ten cross-validation repeats but small, non-uniform across tissues and unvalidated by any formal test. Overall predictive power is weak: out-of-sample R^2 is near zero for every model, closely reproducing the strongest published negative benchmark in the field.

Three further analyses of the same cohort tested how robust that advantage is. A random forest did worse than the linear offset model; on transformed outcome scales the advantage kept its direction but nearly vanished on one; and an expression-only model transferred to unseen tissues with the only clearly positive out-of-sample R^2 of the project, but lost to the global mean when Pancreas or Esophagus/Stomach was held out.

The compound module produced a curated, fully referenced evidence base on *Moringa* constituents in which the extraction solvent is tracked explicitly, and a shortlist of 18 candidates with 5 in tier A, in which the prespecified liability cap and the not-assessable rule both visibly bound. It established that niazimicin is the only constituent confirmed in a dichloromethane fraction of seed and leaf, and that moringin, the isothiocyanate most characteristic of the plant, has no published ionising-radiation study at all.

What remains hypothetical. Everything downstream of the prediction. The pathway ranking is capped at an exploratory tier because no prediction gain was formally shown and no negative controls were run. The cell-line discrepancies are review signals dominated by shrinkage. No compound has been shown to be present in the extracts that were tested, no compound has been shown in this work to modify clonogenic survival, and no claim of selective radioprotection is made or supported. The candidate tiers describe the state of the published evidence, not expected experimental outcomes.

Contribution. The project's contribution is methodological and evidential rather than predictive: a transparent framework; an honest, negative-leaning benchmark that agrees with the best prior work; the correction of its own earlier artefact; an explicit record of a deadline-driven protocol deviation; and a traceable path from the original radiobiology to a small set of experiments that would settle the question. Its most valuable single output may be the mismatch revealed by the compound curation — the compound present in the right solvent has no radiation data, and the compounds with radiation data were never found in that solvent — because it turns a diffuse phytochemical question into two

specific and feasible experiments: characterise the original extracts, and test the best-documented candidates on clonogenic survival in matched normal and tumour cells.

Appendices

Appendix A Supplementary results of the primary benchmark

Table S1. Paired, component-weighted ΔMAE (mSF2 units) in each cross-validation repeat. Repeats re-split the same lines, so their spread is split-to-split variability, not sampling uncertainty. Source: `results/tables/primary_v1/publication_v1/table_s1_repeat_delta_mae.csv`.

Repeat	M2 vs M1 (primary)	M2P vs M1P (sensitivity)
1	0.006258	0.004976
2	0.003042	0.003706
3	0.005528	0.004016
4	0.002201	0.002335
5	0.002619	0.002303
6	0.005018	0.004040
7	−0.002404	0.000379
8	0.007638	0.007469
9	0.005673	0.005690
10	0.002742	0.003793
Mean (pooled estimate)	0.003832	0.003871
Repeats with ΔMAE > 0	9	10

Table S2. Within-tissue Spearman correlation between observed and pooled out-of-fold predicted mSF2, corrected version, for the nine tissues with at least ten components (the eleven smaller tissues are not reported). *Undefined*: the prediction is constant within this tissue in every fitted model, so there is no within-tissue ranking. Defined values are biased downward, because pooled out-of-fold predictions contain each fold's training tissue mean, which is lower when the held-out line's own value is high; they are descriptive only. Source: `results/tables/primary_v1/publication_v1/within_tissue_spearman_v2.csv`.

Tissue	Components	M0	M1	M1P	M2	M2P
Bladder/Urinary Tract	11	*undefined*	*undefined*	−0.227	0.182	0.182
Bowel	21	*undefined*	*undefined*	−0.263	−0.035	−0.047
Breast	17	*undefined*	*undefined*	−0.507	0.252	0.223
CNS/Brain	23	*undefined*	*undefined*	−0.046	0.333	0.333
Esophagus/Stomach	23	*undefined*	*undefined*	−0.294	−0.179	−0.211
Head and Neck	18	*undefined*	*undefined*	0.252	0.153	0.159
Lung	46	*undefined*	*undefined*	−0.118	−0.038	−0.032
Ovary/Fallopian Tube	10	*undefined*	*undefined*	−0.709	0.200	0.200
Pancreas	16	*undefined*	*undefined*	−0.156	−0.262	−0.274

Appendix B Supplementary results of the explainability analysis

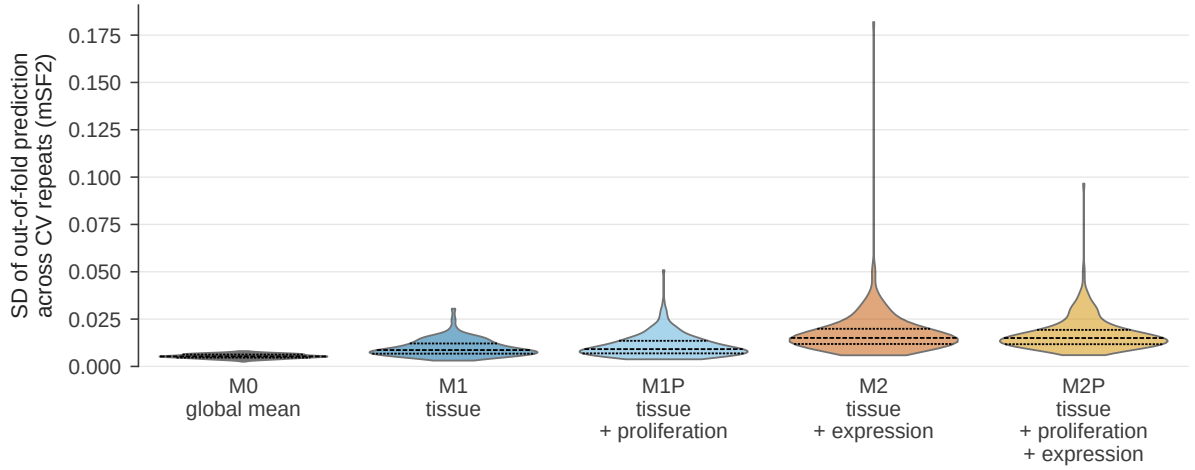


Figure S1. Stability of each cell line's out-of-fold prediction across the ten cross-validation repeats, by model: the distribution over records of the standard deviation of the prediction. Violins show the kernel density over all 241 records with the quartiles marked. The quantity is split-to-split variability of the prediction, not a prediction or confidence interval, and it widens with the number of repeats. The expression models vary more between splits than the tissue-only models, whose prediction is a shrunken tissue mean.

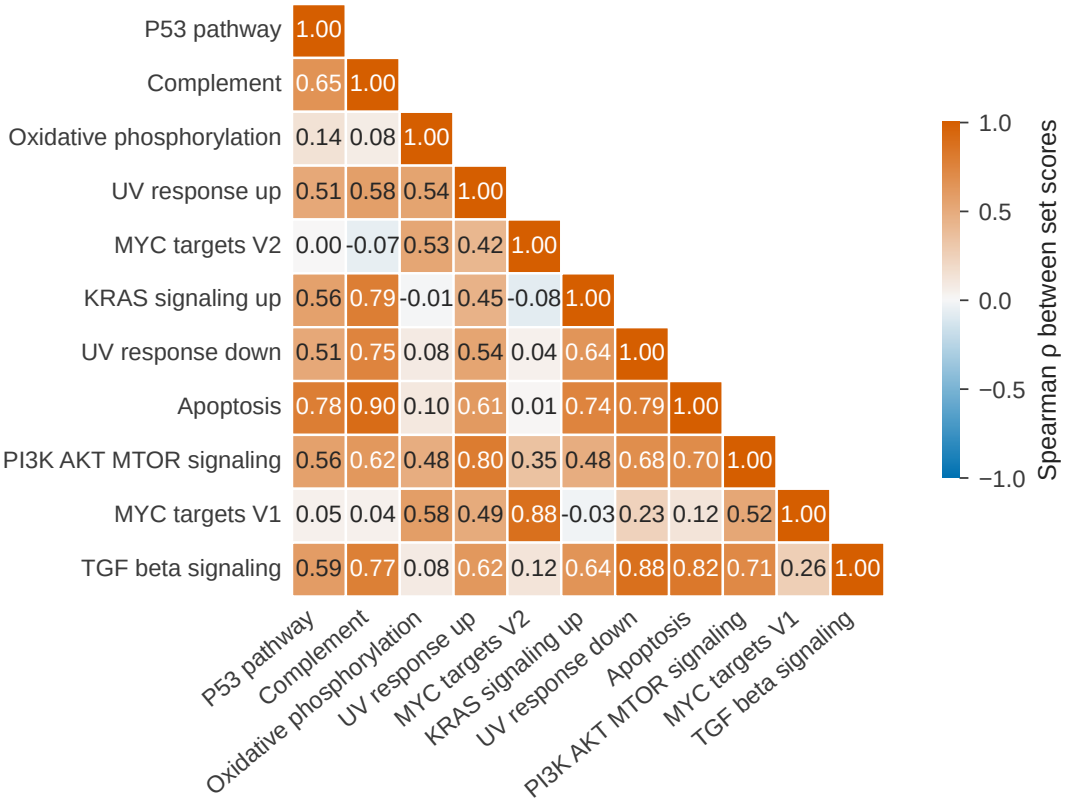


Figure S2. Spearman correlation between the Hallmark gene-set scores of the tier-C sets and of every set flagged as sharing signal with them (absolute correlation at least 0.80), computed over the eligible cell lines from the expression features alone. Rows and columns follow the triage rank order, with no clustering, so no grouping should be read as a biological relationship. Correlated scores can share or exchange permutation importance, so neighbouring sets in Figure 6 are not independent findings.

Table S3. Exploratory Hallmark gene-set ranking with prespecified triage tiers (prefix `HALLMARK_` omitted). Importance is the increase in component-weighted held-out MAE when a score is permuted, averaged over the 50 fits of M2, in units of 10^{-3} mSF2. Shares are proportions of cross-validation fits, not probabilities. Tiers A and B are closed (no formally shown gain; zero null draws); negative-control evidence is not available for any set. A gene-set-level association inside a predictive model is not a gene-level finding, a validated biomarker, a causal mechanism or a radiation-response pathway. Source: `results/tables/primary_v1/publication_triage_v1/table_s2_gene_set_ranking.csv`.

Rank	Gene set	Mean imp.	SD	Share > 0	Sign cons.	Sign	Share zero	Tier	Score co
1	P53_PATHWAY	3.428	3.017	0.90	0.92	negative	0.08	C	—
2	COMPLEMENT	1.181	2.479	0.68	0.92	positive	0.08	C	APOPTO
3	OXIDATIVE_PHOSPHORYLATION	1.070	1.639	0.74	0.86	negative	0.14	C	—
4	UV_RESPONSE_UP	1.000	2.159	0.68	0.76	negative	0.24	C	PI3K_AK NALING
5	MYC_TARGETS_V2	0.638	1.150	0.64	0.88	negative	0.12	C	MYC_TA
6	KRAS_SIGNALING_UP	0.620	1.577	0.58	0.82	positive	0.18	C	—
7	UV_RESPONSE_DN	0.551	1.558	0.60	0.82	positive	0.18	C	TGF_BET
8	EPITHELIAL_MESENCHYMAL_TRANSITION	0.393	1.118	0.38	0.56	positive	0.44	—	—
9	NOTCH_SIGNALING	0.262	1.119	0.40	0.52	negative	0.48	—	—
10	HYPOXIA	0.077	0.514	0.10	0.04	negative	0.96	—	UV_RESI
11	INTERFERON_ALPHA_RESPONSE	0.052	0.353	0.10	0.04	negative	0.96	—	INTER- FERON_ SPONSE
12	KRAS_SIGNALING_DN	0.038	0.591	0.40	0.66	negative	0.34	—	—
13	G2M_CHECKPOINT	0.011	0.075	0.08	0.02	positive	0.98	—	E2F_TAR
14	ADIPOGENESIS	0.003	0.051	0.12	0.10	negative	0.90	—	FATTY_A
15	TNFA_SIGNALING_VIA_NFKB	0.002	0.013	0.08	0.02	negative	0.98	—	INFLAM SPONSE
16	ANDROGEN_RESPONSE	0.000	0.000	0.06	0.00	none	1.00	—	PROTEIN
17	APOPTOSIS	0.000	0.000	0.06	0.00	none	1.00	—	COMPLE
18	E2F_TARGETS	0.000	0.000	0.06	0.00	none	1.00	—	G2M_CH
19	FATTY_ACID_METABOLISM	0.000	0.000	0.06	0.00	none	1.00	—	ADIPOG
20	GLYCOLYSIS	0.000	0.000	0.06	0.00	none	1.00	—	PROTEIN
21	HEME_METABOLISM	0.000	0.000	0.06	0.00	none	1.00	—	PROTEIN
22	IL2_STAT5_SIGNALING	0.000	0.000	0.06	0.00	none	1.00	—	APOPTO
23	IL6_JAK_STAT3_SIGNALING	0.000	0.000	0.06	0.00	none	1.00	—	INFLAM SPONSE
24	INTERFERON_GAMMA_RESPONSE	0.000	0.000	0.06	0.00	none	1.00	—	INTERFE PHA_RES
25	MITOTIC_SPINDLE	0.000	0.000	0.06	0.00	none	1.00	—	G2M_CH
26	MTORC1_SIGNALING	0.000	0.000	0.06	0.00	none	1.00	—	UNFOLD TEIN_RE
27	MYC_TARGETS_V1	0.000	0.000	0.06	0.00	none	1.00	—	MYC_TA
28	PEROXISOME	0.000	0.000	0.06	0.00	none	1.00	—	BILE_AC
29	PI3K_AKT_MTOR_SIGNALING	0.000	0.000	0.06	0.00	none	1.00	—	MITOTIC

Continue

Table S3 (continued)

Rank	Gene set	Mean imp.	SD	Share > 0	Sign cons.	Sign	Share zero	Tier	Score co
30	SPERMATOGENESIS	0.000	0.000	0.06	0.00	none	1.00	—	—
31	TGF_BETA_SIGNALING	0.000	0.000	0.06	0.00	none	1.00	—	UV_RESI
32	UNFOLDED_PROTEIN_RESPONSE	0.000	0.000	0.06	0.00	none	1.00	—	MTORC1
33	ESTROGEN_RESPONSE_EARLY	−0.002	0.024	0.08	0.04	negative	0.96	—	ESTROG SPONSE_
34	CHOLESTEROL_HOMEOSTASIS	−0.005	0.033	0.06	0.02	positive	0.98	—	—
35	ANGIOGENESIS	−0.008	0.124	0.08	0.04	positive	0.94	—	COAGUL
36	APICAL_JUNCTION	−0.008	0.059	0.06	0.02	positive	0.98	—	TGF_BET
37	MYOGENESIS	−0.013	0.429	0.06	0.04	negative	0.96	—	APICAL_
38	PANCREAS_BETA_CELLS	−0.014	0.070	0.06	0.02	none	0.96	—	—
39	INFLAMMATORY_RESPONSE	−0.015	0.107	0.08	0.12	positive	0.88	—	TNFA_SI ING_VIA
40	ALLOGRAFT_REJECTION	−0.026	0.184	0.06	0.02	positive	0.98	—	APOPTO
41	WNT_BETA_CATENIN_SIGNALING	−0.028	0.198	0.12	0.20	negative	0.80	—	—
42	ESTROGEN_RESPONSE_LATE	−0.036	0.202	0.06	0.04	positive	0.96	—	ESTROG SPONSE_
43	BILE_ACID_METABOLISM	−0.037	0.264	0.06	0.02	positive	0.98	—	PEROXIS
44	XENOBIOTIC_METABOLISM	−0.041	0.232	0.06	0.04	positive	0.96	—	—
45	REACTIVE_OXYGEN_SPECIES_PATHWAY	−0.042	0.426	0.18	0.24	positive	0.76	—	—
46	HEDGEHOG_SIGNALING	−0.049	0.271	0.06	0.08	positive	0.92	—	—
47	DNA_REPAIR	−0.054	0.379	0.06	0.02	negative	0.98	—	G2M_CH
48	APICAL_SURFACE	−0.066	0.290	0.06	0.06	positive	0.94	—	—
49	COAGULATION	−0.082	0.461	0.08	0.08	positive	0.92	—	COMPLE
50	PROTEIN_SECRETION	−0.107	0.754	0.06	0.02	positive	0.98	—	PI3K_AK NALING

Note. Tier: C, exploratory tier C; —, not prioritised. Sign: dominant sign of the coefficient across fits. Share zero: share of fits in which the set received a zero coefficient. Values of 0.000 are numerically zero to three decimals (sets never selected). 'Score correlated with': a set whose score correlates with this one at $|\rho| \geq 0.80$, so that importance may be shared or split between the two.

Table S4. Cell-line discrepancy review: every line in discrepancy tiers A, B and C, and MCF-7 (not prioritised), shown for context; the other 192 lines are not prioritised. Predictions are the repeat-averaged out-of-fold predictions (mSF2); disagreement is |M1−M2|; SD is the spread of the M2 prediction across repeats. Tier thresholds are cohort percentiles of the absolute error (Section 4.9). Measured lines are not independent validation samples, and context lines are labelled only; their tiers are not changed. Source: `results/tables/primary_v1/publication_triage_v1/table_s3_discrepancy_review.csv`.

Rank	DepMap ID	Cell line	Tissue	Observed	Pred. M1	Pred. M2	Disagr.	SD M2	Tier	Triage category	Technical flags
1	ACH-000322	HT-144	Skin	0.080	0.594	0.593	0.001	0.013	A	technical review first	small tissue group
2	ACH-000622	KNS-42	CNS/Brain	0.125	0.586	0.596	0.009	0.017	A	unexplained discrepancy	—
3	ACH-000941	HEC-1-B	Uterus	0.970	0.568	0.561	0.006	0.009	A	technical review first	related lines in the cohort; small tissue group
4	ACH-000055	D283 Med	CNS/Brain	0.180	0.585	0.582	0.003	0.023	A	unexplained discrepancy	—
5	ACH-000900	NCI-H23	Lung	0.190	0.593	0.583	0.010	0.012	A	unexplained discrepancy	—
6	ACH-000808	HuH28	Biliary Tract	0.998	0.583	0.625	0.042	0.014	A	technical review first	small tissue group
7	ACH-000048	TOV-112D	Ovary/Fallopian Tube	0.970	0.584	0.546	0.038	0.019	A	unexplained discrepancy	—
8	ACH-000736	SNU-601	Esophagus/Stomach	0.190	0.582	0.560	0.022	0.010	A	unexplained discrepancy	—
9	ACH-000335	MSTO-211H	Pleura	0.970	0.591	0.579	0.012	0.013	A	technical review first	small tissue group
10	ACH-000934	MDA-MB-361	Breast	0.180	0.582	0.542	0.040	0.012	A	unexplained discrepancy	—
11	ACH-000408	TE-5	Esophagus/Stomach	0.211	0.582	0.617	0.035	0.010	A	unexplained discrepancy	—
12	ACH-000657	A2780	Ovary/Fallopian Tube	0.230	0.617	0.577	0.040	0.013	A	unexplained discrepancy	—
13	ACH-000631	KS-1	CNS/Brain	0.233	0.582	0.605	0.022	0.022	B	unexplained discrepancy	—
14	ACH-000149	SK-N-SH	Peripheral Nervous System	0.930	0.584	0.602	0.018	0.013	B	technical review first	related lines in the cohort; Cellosaurus problem flag on a relative (CVCL_0019); small tissue group
15	ACH-000025	CH-157MN	CNS/Brain	0.960	0.575	0.639	0.064	0.026	B	unexplained discrepancy	—
16	ACH-000545	VM-CUB1	Bladder/Urinary Tract	0.920	0.586	0.608	0.022	0.016	B	unexplained discrepancy	—
17	ACH-000520	59M	Ovary/Fallopian Tube	0.910	0.579	0.611	0.032	0.012	B	unexplained discrepancy	—

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Table S4 (continued)

Rank	DepMap ID	Cell line	Tissue	Observed	Pred. M1	Pred. M2	Disagr.	SD M2	Tier	Triage category	Technical flags
18	ACH-000662	COR-L23	Lung	0.291	0.594	0.601	0.007	0.012	B	unexplained discrepancy	—
19	ACH-000423	SK-MEL-3	Skin	0.880	0.578	0.539	0.039	0.023	B	technical review first	small tissue group
20	ACH-000958	SW48	Bowel	0.280	0.584	0.563	0.021	0.018	B	unexplained discrepancy	—
21	ACH-000652	SUIT-2	Pancreas	0.885	0.584	0.562	0.022	0.006	B	unexplained discrepancy	—
22	ACH-000758	MKN74	Esophagus/Stomach	0.293	0.582	0.563	0.019	0.030	C	exploratory candidate	—
23	ACH-000046	ACHN	Kidney	0.861	0.579	0.586	0.006	0.013	C	technical review first	small tissue group
24	ACH-000511	Calu-1	Lung	0.896	0.586	0.626	0.040	0.017	C	exploratory candidate	—
25	ACH-000984	HEC-6	Uterus	0.249	0.572	0.518	0.054	0.018	C	technical review first	small tissue group
26	ACH-000863	DBTRG-05MG	CNS/Brain	0.311	0.580	0.632	0.052	0.011	C	exploratory candidate	—
27	ACH-000538	HUTU80	Bowel	0.278	0.582	0.541	0.041	0.015	C	exploratory candidate	—
28	ACH-000292	NCI-H841	Lung	0.850	0.589	0.559	0.030	0.027	C	exploratory candidate	—
29	ACH-000903	FTC-133	Thyroid	0.338	0.589	0.613	0.024	0.013	C	technical review first	small tissue group
30	ACH-000940	AN3 CA	Uterus	0.360	0.576	0.526	0.051	0.035	C	technical review first	small tissue group; prediction unstable across CV repeats
31	ACH-000355	NCI-H82	Lung	0.730	0.589	0.517	0.071	0.020	C	technical review first	M1 and M2 disagree strongly
32	ACH-000470	SW1463	Bowel	0.685	0.571	0.534	0.037	0.036	C	technical review first	prediction unstable across CV repeats
33	ACH-000358	NCI-H69	Lung	0.425	0.593	0.543	0.050	0.038	C	technical review first	prediction unstable across CV repeats
34	ACH-000169	RD	Soft Tissue	0.690	0.589	0.494	0.095	0.182	C	technical review first	small tissue group; prediction unstable across CV repeats; M1 and M2 disagree strongly
35	ACH-000544	OE21	Esophagus/Stomach	0.710	0.567	0.611	0.044	0.033	C	technical review first	prediction unstable across CV repeats
36	ACH-000464	CAS-1	CNS/Brain	0.740	0.575	0.659	0.083	0.040	C	technical review first	prediction unstable across CV repeats; M1 and M2 disagree strongly
37	ACH-001306	8305C	Thyroid	0.736	0.582	0.658	0.075	0.015	C	technical review first	small tissue group; M1 and M2 disagree strongly
38	ACH-000703	DMS 79	Lung	0.507	0.593	0.579	0.014	0.035	C	technical review first	prediction unstable across CV repeats
39	ACH-000480	HuH-7	Liver	0.510	0.584	0.652	0.067	0.051	C	technical review first	small tissue group; prediction unstable across CV repeats; M1 and M2 disagree strongly
40	ACH-000979	DU 145†	Prostate	0.710	0.580	0.650	0.071	0.048	C	technical review first	stranded RNA profile; small tissue group; prediction unstable across CV repeats; M1 and M2 disagree strongly
41	ACH-000771	BICR 56	Head and Neck	0.650	0.599	0.703	0.103	0.021	C	technical review first	M1 and M2 disagree strongly
42	ACH-000244	DK-MG	CNS/Brain	0.529	0.574	0.667	0.094	0.030	C	technical review first	M1 and M2 disagree strongly

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Table S4 (continued)

Rank	DepMap ID	Cell line	Tissue	Observed	Pred. M1	Pred. M2	Disagr.	SD M2	Tier	Triage category	Technical flags
43	ACH-000415	BICR 6	Head and Neck	0.714	0.594	0.683	0.089	0.034	C	technical review first	prediction unstable across CV repeats; M1 and M2 disagree strongly
44	ACH-000665	SK-MES-1	Lung	0.675	0.583	0.647	0.065	0.033	C	technical review first	prediction unstable across CV repeats
45	ACH-000138	CFPAC-1	Pancreas	0.554	0.588	0.668	0.079	0.017	C	technical review first	M1 and M2 disagree strongly
46	ACH-000254	SCC-15	Head and Neck	0.585	0.598	0.670	0.072	0.017	C	technical review first	M1 and M2 disagree strongly
47	ACH-000148	Hs 578T	Breast	0.650	0.566	0.644	0.078	0.050	C	technical review first	prediction unstable across CV repeats; M1 and M2 disagree strongly
48	ACH-000188	SCC-25	Head and Neck	0.675	0.598	0.680	0.082	0.036	C	technical review first	prediction unstable across CV repeats; M1 and M2 disagree strongly
206	ACH-000019	MCF7†	Breast	0.520	0.574	0.534	0.040	0.016	—	not prioritised	—

Note. † Context line from the motivating *Moringa* study (DU145 and MCF-7). 'Unexplained discrepancy': unexplained discrepancy requiring follow-up (tier A or B without technical flags); 'exploratory candidate': tier C without technical flags; 'technical review first': at least one technical flag. Rank 206 of 241 for MCF-7.

Appendix C Compound evidence records

The complete curated records, with every citation, DOI, verification route, reading depth, extraction solvent, identification method, model system, radiation dose, concentration, measured genes and reported direction, are held in `data/compounds/moringa_constituents.yaml` and `data/compounds/compound_biology_evidence.yaml`; the curation audit trail (searches, exclusions, identifier problems and unresolved gaps) is in `docs/compound_evidence_curation.md`. Tables S5–S8 present the evidence used for the ranking. DOIs of the primary studies are listed in the tables; only the studies discussed in the text appear in the reference list.

Table S5. Inventory of documented *M. oleifera* seed and leaf constituents: one row per primary analytical report, with the extraction solvent or fraction as reported and the strength of identification (isolation and NMR or reference standard; spectral-library match; not stated). DCM: the report concerns a dichloromethane (methylene chloride) extract or fraction. Benzyl isothiocyanate is listed because it was evaluated, although its only report concerns its precursor in roots. Source: `results/tables/primary_v1/compounds_v1/table_c1_constituent_inventory.csv`.

Compound	#	Part	Solvent or fraction	DCM	Identification method	Strength	Research group	DOI
Acetylated moringin[a]	1	leaf	Ethanol leaf extract	no	Isolation; structure elucidated by NMR	isolation, NMR	Faizi / Siddiqui (Karachi)	10.1021/np50111a011
	2	leaf	Fresh leaves extracted with water	no	Not stated in the abstract	not stated	Raskin (Rutgers)	10.1016/j.phytochem.2014.03.028
α-Tocopherol	1	leaf	Leaf supercritical CO_2 extract	no	GC-MS with NIST library and Kovats retention indices, quantified against a calibration curve	reference standard	Kessler (2025)	10.1111/jocd.70486
	2	seed	Seed oil (cold press, *n*-hexane, or chloroform/methanol 50:50)	no	Tocopherols by HPLC (α, γ and δ reported)	not stated	Tsaknis / Lalas	10.1021/jf9904214
Astragalin	1	leaf	Not stated in the abstract (full text paywalled)	no	HPLC and GC-MS per MeSH; criteria not verified	not stated	Bennett / Mellon (Norwich)	10.1021/jf0211480
	2	leaf	Leaf water extract	no	Standard purchased for quantitation; identification against literature MS data; recorded conservatively as a library match	library match	Wang J (2024)	10.1016/j.fochx.2024.101585
Benzyl isothiocyanate	1	other	Roots (precursor benzyl glucosinolate), not seed or leaf	no	HPLC and GC-MS per MeSH	not stated	Bennett / Mellon (Norwich)	10.1021/jf0211480
β-Sitosterol	1	seed	Ethanol seed extract	no	Isolation (also β-sitosterol-3-*O*-glucoside and an oleoyl-glucoside); method not stated	not stated	Guevara / Nishino	10.1016/s1383-5718(99)00025-x
	2	seed	Seed oil (cold press, *n*-hexane, or chloroform/methanol 50:50)	no	Sterol analysis; up to 50.07% of the sterol fraction	not stated	Tsaknis / Lalas	10.1021/jf9904214
	3	seed	Bio-guided column fractionation; solvent not stated	no	GC-MS	library match	Raafat (Beirut)	10.1007/s11655-017-2758-4
Chlorogenic acid	1	leaf	Not stated in the abstract (full text paywalled)	no	HPLC and GC-MS per MeSH; 3- and 5-caffeoylquinic acids reported	not stated	Bennett / Mellon (Norwich)	10.1021/jf0211480
	2	leaf	Leaf water extract	no	Standard purchased for quantitation; identification against literature MS data; recorded as a library match	library match	Wang J (2024)	10.1016/j.fochx.2024.101585

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Table S5 (continued)

Compound	#	Part	Solvent or fraction	DCM	Identification method	Strength	Research group	DOI
cis-Vaccenic acid	1	leaf	Dichloromethane leaf extract	**yes**	GC-MS; 10.68% of the reported profile; criteria not verified	library match	Khurana (Amity)	10.5958/0975-6892.2019.00004.2
O-Ethyl carbamate[b]	1	seed	Ethanol seed extract	no	Reported as a new compound; method not stated	not stated	Guevara / Nishino	10.1016/s1383-5718(99)00025-x
	2	seed	85% ethanol seed extract, D101 resin chromatography	no	Isolated as compound 7; data compared with the literature	isolation, NMR	Li LZ	10.3390/molecules28176426
Glucomoringin	1	both	Not stated in the abstract (full text paywalled)	no	HPLC and GC-MS per MeSH; criteria not verified	not stated	Bennett / Mellon (Norwich)	10.1021/jf0211480
	2	both	Not stated in the abstract	no	Not stated in the abstract	not stated	Bennett / Mellon (Norwich)	10.1016/j.foodchem.2010.03.073
	3	seed	85% ethanol seed extract, D101 resin chromatography	no	Isolated as compound 16; data compared with the literature	isolation, NMR	Li LZ	10.3390/molecules28176426
	4	leaf	Leaf water reflux extract	no	Orbitrap MS/MS fragmentation compared with the literature; no authentic standard	library match	Wang J (2024)	10.1016/j.fochx.2024.101585
4-Hydroxyphenyl-acetonitrile	1	leaf	Dichloromethane fraction of a leaf methanolic extract	**yes**	GC-MS; reported as a major compound; full list and criteria not verified (paywalled)	library match	Arsad (USM)	10.1007/s11033-021-06466-y
	2	seed	Ripe seed: crude 80% ethanol extract, butanol fraction and aqueous residue (DCM fraction prepared, GC-MS not reported)	no	GC-MS with NIST library match > 80%; no authentic standards	library match	Arsad (USM)	10.4103/pm.pm_212_17
Isoquercitrin	1	leaf	Not stated in the abstract (full text paywalled)	no	HPLC and GC-MS per MeSH; also quercetin-3-*O*-(6″-malonyl)glucoside	not stated	Bennett / Mellon (Norwich)	10.1021/jf0211480
	2	leaf	Leaf water extract	no	Standard purchased for quantitation; identification by formula and literature MS; recorded as a library match	library match	Wang J (2024)	10.1016/j.fochx.2024.101585
Kaempferol	1	seed	Not stated in the abstract	no	LC-HRMS dereplication (tentative annotation, no standard)	library match	Shady (2022)	10.3390/antiox11091743
Methyl 4-hydroxyphenylacetate	1	leaf	Dichloromethane fraction of a leaf methanolic extract	**yes**	GC-MS; reported as a major compound; criteria not verified	library match	Arsad (USM)	10.1007/s11033-021-06466-y
Moringin	1	seed	Ethanol seed extract	no	Isolation; method not stated	not stated	Guevara / Nishino	10.1016/s1383-5718(99)00025-x
	2	leaf	Fresh leaves extracted with water	no	Not stated (full text not retrieved)	not stated	Raskin (Rutgers)	10.1016/j.phytochem.2014.03.028

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Table S5 (continued)

Compound	#	Part	Solvent or fraction	DCM	Identification method	Strength	Research group	DOI
	3	leaf	Leaf fast centrifugal partition chromatography fractions	no	Not stated in the abstract	not stated	Raskin (Rutgers)	10.1021/jf505014n
Niazimicin	1	both	CH_2Cl_2 partition fraction of a 70% ethanol maceration of seeds and of leaves	**yes**	Reference standard isolated from the seed CH_2Cl_2 fraction (purity 96.6% by PDA); identity by HRMS, UV, IR, EI-MS and 1D/2D NMR; quantified by validated UPLC-MS/MS (MRM 358 → 107)	isolation, NMR	Hanafi / Abdel Motaal (Cairo)	10.38212/2224-6614.3282
	2	seed	Methylene-chloride and ethyl-acetate fractions of a seed extract	**yes**	Not stated in the abstract	not stated	Hanafi / Abdel Motaal (Cairo)	10.1111/jfbc.13992
	3	seed	Ethanol seed extract	no	Isolation; method not stated	not stated	Guevara / Nishino	10.1016/s1383-5718(99)00025-x
	4	seed	Ethyl-acetate fraction of an ethanolic seed extract	no	Chromatographic isolation	not stated	Albuquerque Costa (Ceará)	10.1155/2017/7963747
	5	seed	85% ethanol seed extract, D101 resin chromatography	no	Isolated as compound 11; spectroscopic data compared with the literature	isolation, NMR	Li LZ	10.3390/molecules28176426
Niaziminin	1	leaf	Ethanol leaf extract	no	Isolation; structure elucidated by NMR	isolation, NMR	Faizi / Siddiqui (Karachi)	10.1021/np50111a011
	2	leaf	Leaf; solvent not given in the abstract	no	Not stated in the abstract	not stated	Ohigashi (Kyoto)	10.1055/s-2006-957442
Niazirin	1	leaf	Ethanol leaf extract	no	Bioassay-guided isolation; spectroscopy including 2D NMR	isolation, NMR	Faizi / Siddiqui (Karachi)	10.1021/np50111a011
	2	seed	Ethanol seed extract	no	Isolation; method not stated	not stated	Guevara / Nishino	10.1016/s1383-5718(99)00025-x
	3	seed	85% ethanol seed extract, D101 resin chromatography	no	Isolated as compound 9; data compared with the literature	isolation, NMR	Li LZ	10.3390/molecules28176426
Oleic acid	1	seed	Seed oil (cold press, *n*-hexane, or chloroform/methanol 50:50)	no	Fatty-acid methyl-ester analysis; up to 75.39% of the oil	not stated	Tsaknis / Lalas	10.1021/jf9904214
Palmitic acid	1	leaf	Dichloromethane leaf extract	**yes**	GC-MS; 96 compounds reported, *n*-hexadecanoic acid the largest peak (28.35%); library matching assumed but not verified	library match	Khurana (Amity)	10.5958/0975-6892.2019.00004.2
	2	seed	Seed oil (cold press, *n*-hexane, or chloroform/methanol 50:50)	no	Fatty-acid methyl-ester analysis	not stated	Tsaknis / Lalas	10.1021/jf9904214
Phytol	1	leaf	Hydroethanolic leaf extract	no	GC-MS; 23.54% of the leaf profile; method details not verified	library match	Vats / Gupta (Rajasthan)	10.1007/s12298-016-0407-6

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Table S5 (continued)

Compound	#	Part	Solvent or fraction	DCM	Identification method	Strength	Research group	DOI
Quercetin	1	seed	Not stated in the abstract	no	LC-HRMS dereplication (tentative annotation, no standard)	library match	Shady (2022)	10.3390/antiox11091743
	2	leaf	Leaf water extract	no	Listed in the constituent table; no authentic standard	library match	Wang J (2024)	10.1016/j.fochx.2024.101585
Rutin	1	leaf	*M. oleifera* leaf water extract	no	Standard purchased for quantitation; identification against literature MS data; recorded as a library match	library match	Wang J (2024)	10.1016/j.fochx.2024.101585
	2	leaf	Not stated in the abstract	no	Quercetin rutinosides among the leaf flavonoids; criteria not stated	not stated	Bennett / Mellon (Norwich)	10.1016/j.foodchem.2010.03.073

Note. [a] 4-[(4′-*O*-Acetyl-*α*-L-rhamnosyloxy)benzyl] isothiocyanate. [b] *O*-Ethyl-4-(*α*-L-rhamnosyloxy)benzyl carbamate. Research groups are those used to count independent sources; reports from one group count once (for example the two niazimicin reports of the Cairo group, and the two 4-hydroxyphenylacetonitrile reports of the USM group). Identity caveats recorded in the source file include: the glucomoringin identifier is ambiguous (every PubChem record is a potassium salt with a different stereo-InChIKey); free quercetin and kaempferol may arise by hydrolysis of glycosides; the C18:1 isomer of the *cis*-vaccenic acid report cannot be settled by library matching; and niazirin was explicitly *not* identified in the dichloromethane fractions analysed by **abdelsayed2021jfda**.

Table S6. Primary evidence linking the constituents to radiation-response biology. Level L3: ionising-radiation study of the purified compound; L2: purified compound modulating a relevant process without radiation. Reported directions are never merged: protection of normal cells, protection of tumour cells, sensitisation of tumour cells, process modulation without radiation, genotoxicity or no effect. Liability: evidence that caps the tier at C. Scored: the record counts towards the score (purified compound). Source: `results/tables/primary_v1/compounds_v1/table_c2_biology_evidence.csv`.

ID	Compound	Level	Process	Model system	Radiation	Dose (Gy)	Concentration	Direction	Liab.	Scored	DOI
TOC-01	α-Tocopherol	L3	proliferation	Rats (Sprague-Dawley), exteriorised mid-small bowel	X-rays	11	not stated (luminal phosphate and acetate esters; also dietary)	protection, normal	no	yes	10.1097/00000658-199522240-00008
TOC-02	α-Tocopherol	L3	other	Humans, $n = 54$, oral cavity and oropharyngeal cancer patients	External-beam radiotherapy	2 per fraction over 5–7 weeks; total not stated	not stated (oil-solution rinse)	protection, normal	no	yes	10.1002/hed.10382
AST-01	Astragalin	L2	DNA damage	*In vitro* DNA and *Saccharomyces cerevisiae*	none	not stated	not stated	genotoxic	**yes**	yes	10.3390/ijms13032846
BIT-01	Benzyl isothiocyanate	L3	cell cycle	BxPC-3 human pancreatic cancer	Gamma	5	2.5 µM	sensitisation, tumour	no	yes	10.2741/e55
BIT-02	Benzyl isothiocyanate	L3	apoptosis	PANC-1 and MIAPaCa-2 human pancreatic cancer	X-rays	not stated	5–10 µM (PANC-1); 2.5–5 µM (MIAPaCa-2)	sensitisation, tumour	no	yes	10.3892/ijmm.2011.770
BST-01	β-Sitosterol	L3	DNA damage	Mouse thymocytes (BALB/c)	Ionising; modality unverified	6	not stated	protection, normal	no	yes	10.1002/jcb.21326
BST-02	β-Sitosterol	L3	oxidative stress	Adjuvant-arthritic male rats with and without irradiation	Gamma	2 per week × 4 (8 total)	40 mg/kg/day oral	protection, normal	no	yes	10.1080/08923973.2025.2591423
CGA-01	Chlorogenic acid	L3	oxidative stress	Hepatocellular carcinoma cell lines (not stated) and a murine model	Ionising	not stated	not stated	**protection, tumour**	**yes**	yes	10.1155/2022/4566949
CGA-02	Chlorogenic acid	L3	DNA damage	Mouse bone marrow	Gamma, whole body	1.15 (0.05 Gy/s)	50, 100, 200 mg/kg oral	protection, normal	no	yes	10.1016/0165-7992(93)90022-n
CGA-03	Chlorogenic acid	L3	apoptosis	Mice, $n = 70$, testis	Ionising, total body	2.0	5–80 mg/kg i.p.	protection, normal	no	yes	10.1016/j.heliyon.2022.e10798
KAE-01	Kaempferol	L3	cell cycle	A-549 lung cancer and HFL1 normal lung fibroblasts; nude-mouse xenograft	Ionising	not stated	not stated	sensitisation, tumour	no	yes	10.3892/or.2015.4204
KAE-02	Kaempferol	L3	apoptosis	C57BL/6 mice and HUVECs	Gamma	8.5 and 7 total body; *in vitro* not stated	not stated	protection, normal	no	yes	10.1016/j.etap.2018.04.014
MOR-01	Moringin	L2	apoptosis	SH-SY5Y human neuroblastoma	none	—	16.4 µM	process, no radiation	no	yes	10.3390/ijms20081930

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Table S6 (continued)

ID	Compound	Level	Process	Model system	Radiation	Dose (Gy)	Concentration	Direction	Liab.	Scored	DOI
MOR-02	Moringin	L2	cell cycle	PC-3 human prostate cancer	none	—	IC_{50} 3.5 µg/mL	process, no radiation	no	yes	10.3390/molecules28073214
MOR-03	Moringin	L2	oxidative stress	Differentiated NSC-34 motor-neuron-like cells	none	—	0.5, 5, 10 µM (α-cyclodextrin complex)	process, no radiation	no	yes	10.3390/antiox13070813
MOR-04	Moringin	L2	apoptosis	Carcinoma, myeloma and leukaemia lines; nude-mouse xenografts	none	—	micromolar range; not stated	process, no radiation	no	yes	10.1016/j.bcp.2009.12.008
NZM-01	Niazimicin	L2	oxidative stress	Rats, aluminium-chloride dementia model	none	—	250 mg/kg/day (extract fraction)	process, no radiation	no	**no**	10.1111/jfbc.13992
NZR-01	Niazirin	L2	oxidative stress	High-glucose vascular smooth muscle cells; diabetic mice	none	—	not stated	process, no radiation	no	yes	10.1016/j.phymed.2019.153066
OLE-01	Oleic acid	L3	proliferation	Human LDV lymphoid cells	Ionising (air or nitrogen)	not stated	not stated	**no effect**	no	yes	10.1080/09553008314550431
PHY-01	Phytol	L2	apoptosis	A549 human lung carcinoma	none	—	not stated	process, no radiation	no	yes	10.1016/j.biopha.2017.05.066
PHY-02	Phytol	L2	oxidative phosphorylation	A549 human lung carcinoma	none	—	not stated	process, no radiation	no	yes	10.1016/j.biopha.2018.06.035
QUE-01	Quercetin	L3	DNA repair	Human ovarian cancer cells (not stated) and xenograft	X-rays	not stated	not stated	sensitisation, tumour	no	yes	10.2147/ott.s147316
QUE-02	Quercetin	L3	proliferation	Medulloblastoma lines versus neural precursors and normal fibroblasts; orthotopic xenografts	Ionising, fractionated	not stated	low micromolar	sensitisation, tumour	no	yes	10.18632/oncotarget.7980
QUE-03	Quercetin	L3	DNA damage	Human peripheral blood lymphocytes; plasmid DNA	Gamma	1–4 (cells); 6 (plasmid)	3–48 µM; 24 µM optimal	protection, normal	no	yes	10.1016/j.mrgentox.2008.03.003
QUE-04	Quercetin	L3	other	C3H/HeN mouse skin, human and murine fibroblasts, murine tumours	X-rays	35	1% in chow	protection, normal	no	yes	10.1667/rr3237.1
RUT-01	Rutin	L3	DNA damage	Human peripheral blood lymphocytes	Gamma	2	25 µg/mL	protection, normal	no	yes	10.4103/ijnm.ijnm_30_17
RUT-02	Rutin	L3	oxidative stress	Rat brain	Gamma	5	200 mg/kg/day oral	protection, normal	no	yes	10.1080/13813455.2017.1374978

Note. NZM-01 is recorded so that the absence of purified-niazimicin biology is explicit: the dosed material was an extract fraction, not the purified compound, so the record does not score. CGA-01 is liability evidence and is never pooled with the normal-tissue chlorogenic-acid studies. QUE-04 reports no effect on fibroblast or murine tumour radioresponse. Several abstracts omit cell lines, doses or concentrations; these gaps are recorded as ‘not stated’ and must be recovered from the full texts.

Table S7. Written justification of every point of the candidate ranking (Table 12) and of each tier, as recorded by the scoring code. Source: `results/tables/primary_v1/compounds_v1/table_c3_candidate_ranking.csv`.

Compound	D_1 identity	D_2 radiation biology	D_3 independent sources	D_4 feasibility	Tier reason
α-Tocopherol	1 primary report with structural confirmation (reference standard)	Ionising-radiation evidence from 2 independent groups (Delaney, Minnesota; Ferreira, Brazil)	2 groups reporting it in *Moringa*, 2 in the biology evidence, 4 distinct in total	Identifier verified; 2 scored studies report a usable concentration	Total 8/8 with $D_1 \geq 1$ and $D_2 \geq 2$
Chlorogenic acid	2 primary reports; identification by spectral-library match or not stated	Ionising-radiation evidence from 3 independent groups (Abraham / Kesavan; Shoorei / Karimipour; Yan)	2 groups in *Moringa*, 3 in the biology evidence, 5 distinct	Identifier verified; 3 scored studies report a usable concentration	Capped at C from A (total 7/8): liability evidence CGA-01, protection of tumour cells
Kaempferol	1 primary report; spectral-library match or not stated	Ionising-radiation evidence from 2 independent groups (Yao, Taiwan; Yuan, China)	1 group in *Moringa*, 2 in the biology evidence, 3 distinct	Identifier verified; 2 scored studies report a usable concentration	Total 7/8 with $D_1 \geq 1$ and $D_2 \geq 2$
Quercetin	2 primary reports; spectral-library match or not stated	Ionising-radiation evidence from 4 independent groups (Citrin, NCI; Gong / Liu, China; Hulleman / Wurdinger, Amsterdam; Menon, Annamalai)	2 groups in *Moringa*, 4 in the biology evidence, 6 distinct	Identifier verified; 4 scored studies report a usable concentration	Total 7/8 with $D_1 \geq 1$ and $D_2 \geq 2$
Rutin	2 primary reports; spectral-library match or not stated	Ionising-radiation evidence from 2 independent groups (Abdel-Rafei / Thabet, NCRRT Egypt; Patil, Mangalore)	2 groups in *Moringa*, 2 in the biology evidence, 4 distinct	Identifier verified; 2 scored studies report a usable concentration	Total 7/8 with $D_1 \geq 1$ and $D_2 \geq 2$
β-Sitosterol	3 primary reports; spectral-library match or not stated	Ionising-radiation evidence from 2 independent groups (Abdel-Rafei / Thabet, NCRRT Egypt; Wang SQ, China)	3 groups in *Moringa*, 2 in the biology evidence, 5 distinct	Identifier verified; 2 scored studies report a usable concentration	Total 7/8 with $D_1 \geq 1$ and $D_2 \geq 2$
Benzyl isothiocyanate	No primary analytical study identifies it in *Moringa* seed or leaf (review-only, tentative or another plant part)	Ionising-radiation evidence from 2 independent groups (Kubota / Okayasu; Srivastava)	1 group in *Moringa*, 2 in the biology evidence, 3 distinct	Identifier verified; 2 scored studies report a usable concentration	Not assessable: no primary *Moringa* identification (prespecified rule)
Niazirin	2 primary reports with structural confirmation (isolation, NMR)	No ionising-radiation study; 1 study of a relevant process without radiation (Weng, China)	3 groups in *Moringa*, 1 in the biology evidence, 4 distinct	Identifier verified; 1 scored study reports a usable concentration	Total 6/8 with $D_1 \geq 1$ and $D_2 \geq 1$
Oleic acid	1 primary report; spectral-library match or not stated	Ionising-radiation evidence from one group (Cramp, UK)	1 group in *Moringa*, 1 in the biology evidence, 2 distinct	Identifier verified; 1 scored study reports a usable concentration	Total 5/8 with $D_1 \geq 1$ and $D_2 \geq 1$
Astragalin	2 primary reports; spectral-library match or not stated	No ionising-radiation study; 1 study of a relevant process without radiation (Saito)	2 groups in *Moringa*, 1 in the biology evidence, 3 distinct	Identifier verified; 1 scored study reports a usable concentration	Capped at C from B (total 5/8): liability evidence AST-01, genotoxic

Continued on next page

Table S7 (continued)

Compound	D_1 identity	D_2 radiation biology	D_3 independent sources	D_4 feasibility	Tier reason
Moringin	3 primary reports; spectral-library match or not stated	No ionising-radiation study; 4 studies of relevant processes without radiation (Abdull Razis; D'Incalci; Mazzon; Navarra)	2 groups in *Moringa*, 4 in the biology evidence, 6 distinct	Identifier verified; 4 scored studies report a usable concentration	Total 5/8 with $D_1 \geq 1$ and $D_2 \geq 1$
Phytol	1 primary report; spectral-library match or not stated	No ionising-radiation study; 2 studies of relevant processes without radiation (Devi; Thakkar)	1 group in *Moringa*, 2 in the biology evidence, 3 distinct	Identifier verified; 2 scored studies report a usable concentration	Total 5/8 with $D_1 \geq 1$ and $D_2 \geq 1$
Glucomoringin	1 primary report with structural confirmation (isolation, NMR)	No primary study of the purified compound in a relevant process	3 groups in *Moringa*, 0 in the biology evidence, 3 distinct	No concentration reported in any scored study	Total 4/8 with $D_1 \geq 1$ and $D_2 \geq 0$
Niazimicin	2 primary reports with structural confirmation (isolation, NMR)	No primary study of the purified compound in a relevant process	4 groups in *Moringa*, 0 in the biology evidence, 4 distinct	No concentration reported in any scored study	Total 4/8 with $D_1 \geq 1$ and $D_2 \geq 0$
Acetylated moringin	1 primary report with structural confirmation (isolation, NMR)	No primary study of the purified compound in a relevant process	2 groups in *Moringa*, 0 in the biology evidence, 2 distinct	No concentration reported in any scored study	Total 3/8 with $D_1 \geq 1$ and $D_2 \geq 0$
Niaziminin	1 primary report with structural confirmation (isolation, NMR)	No primary study of the purified compound in a relevant process	2 groups in *Moringa*, 0 in the biology evidence, 2 distinct	No concentration reported in any scored study	Total 3/8 with $D_1 \geq 1$ and $D_2 \geq 0$
O-Ethyl carbamate	1 primary report with structural confirmation (isolation, NMR)	No primary study of the purified compound in a relevant process	2 groups in *Moringa*, 0 in the biology evidence, 2 distinct	No concentration reported in any scored study	Total 3/8 with $D_1 \geq 1$ and $D_2 \geq 0$
Isoquercitrin	2 primary reports; spectral-library match or not stated	No primary study of the purified compound in a relevant process	2 groups in *Moringa*, 0 in the biology evidence, 2 distinct	No concentration reported in any scored study	Total 2/8 with $D_1 \geq 1$ and $D_2 \geq 0$
Palmitic acid	2 primary reports; spectral-library match or not stated	No primary study of the purified compound in a relevant process	2 groups in *Moringa*, 0 in the biology evidence, 2 distinct	No concentration reported in any scored study	Total 2/8 with $D_1 \geq 1$ and $D_2 \geq 0$
4-Hydroxyphenyl-acetonitrile	2 primary reports; spectral-library match or not stated	No primary study of the purified compound in a relevant process	At most one independent group (Arsad, USM)	No concentration reported in any scored study	Total 1/8, below the tier-C threshold
Methyl 4-hydroxyphenylacetate	1 primary report; spectral-library match or not stated	No primary study of the purified compound in a relevant process	At most one independent group (Arsad, USM)	No concentration reported in any scored study	Total 1/8, below the tier-C threshold
cis-Vaccenic acid	1 primary report; spectral-library match or not stated	No primary study of the purified compound in a relevant process	At most one independent group (Khurana, Amity)	No concentration reported in any scored study	Total 1/8, below the tier-C threshold

Table S8. Overlap between the genes a published experiment measured for a compound and the members of a tier-C Hallmark gene set (prefix `HALLMARK_` omitted). An overlap marks where a compound and a model-highlighted gene set touch the same genes, which is a hypothesis to test; it is not evidence that the compound modifies the gene set, the pathway or clonogenic SF2, and the gene-set ranking itself is capped at tier C. Benzyl isothiocyanate is not assessable and is not in the shortlist. Source: `results/tables/primary_v1/compounds_v1/table_c4_pathway_overlap.csv`.

Compound	Gene set	Overlap	Measured	In set	Overlapping genes	Evidence
Benzyl isothiocyanate	P53_PATHWAY	2	9	200	APAF1; CDKN1A	BIT-01; BIT-0
	UV_RESPONSE_DN	2	9	144	MAPK14; NFKB1	BIT-01
β-Sitosterol	OXIDATIVE_PHOSPHORYLATION	2	10	200	AIFM1; CYCS	BST-01
	P53_PATHWAY	2	10	200	FOS; JUN	BST-01
	UV_RESPONSE_UP	2	10	158	CASP3; FOS	BST-01
	COMPLEMENT	1	10	200	CASP3	BST-01
Chlorogenic acid	OXIDATIVE_PHOSPHORYLATION	1	3	200	BAX	CGA-03
	P53_PATHWAY	1	3	200	BAX	CGA-03
Kaempferol	COMPLEMENT	2	4	200	CASP3; CASP9	KAE-02
	OXIDATIVE_PHOSPHORYLATION	1	4	200	CYCS	KAE-02
	UV_RESPONSE_UP	1	4	158	CASP3	KAE-02
Moringin	P53_PATHWAY	3	11	200	BAX; CDKN1A; TP53	MOR-01; MOR-02; MOR-03
	COMPLEMENT	2	11	200	CASP3; CASP9	MOR-01; MOR-03
	OXIDATIVE_PHOSPHORYLATION	1	11	200	BAX	MOR-01; MOR-02; MOR-03
	UV_RESPONSE_DN	1	11	144	NFKB1	MOR-01; MOR-04
	UV_RESPONSE_UP	1	11	158	CASP3	MOR-01; MOR-03
Phytol	COMPLEMENT	2	5	200	CASP3; CASP9	PHY-01
	P53_PATHWAY	1	5	200	FAS	PHY-01
	UV_RESPONSE_UP	1	5	158	CASP3	PHY-01
Quercetin	P53_PATHWAY	5	9	200	BAX; CDKN1A; DDIT3; TGFB1; TP53	QUE-01; QUE-04
	OXIDATIVE_PHOSPHORYLATION	1	9	200	BAX	QUE-01
	UV_RESPONSE_UP	1	9	158	H2AX	QUE-01
Rutin	KRAS_SIGNALING_UP	2	7	200	IL1B; NGF	RUT-02
	COMPLEMENT	1	7	200	IL6	RUT-02
	UV_RESPONSE_UP	1	7	158	IL6	RUT-02

Appendix D Data provenance and table correspondence

Dataset manifests are held in `data/manifests/`, one file per source and release, with DOI, URL, licence, access date, per-file SHA-256, column meanings, units, endpoint definition and transformations. The run records are `results/models/primary_v1/run_manifest.json`, `results/tables/primary_v1/ev`

aluation_record.json, results/triage/primary_v1/triage_manifest.json and the compound record results/tables/primary_v1/compounds_v1/compound_record.json; each records the git commit, a flag for uncommitted changes, configuration and file checksums, package versions and UTC timestamps. Table S9 maps every result table of this report to the project's table identifier and source file.

Table S9. Correspondence between the result tables of this report, the project's table identifiers and the saved output files.

This report	Project ID	Source file
Table 4	Table 1	results/tables/primary_v1/publication_v1/table_1_dataset_composition.csv
Table 5	Table 2	results/tables/primary_v1/publication_v1/table_2_model_comparison.csv
Table 7	Table 3	results/tables/primary_v1/publication_v1/table_3_tissue_performance.csv
Table 9	Table R1	results/tables/exploratory_m3_v1/model_metrics.csv
Table 10	Table R2	results/tables/loto_v1/held_out_tissue.csv
Table 11	Table R3	results/tables/scales_v1/comparisons.csv
Table 12	Table C3	results/tables/primary_v1/compounds_v1/table_c3_candidate_ranking.csv
Table S1	Table S1	results/tables/primary_v1/publication_v1/table_s1_repeat_delta_mae.csv
Table S2	Table S4	results/tables/primary_v1/publication_v1/within_tissue_spearman_v2.csv
Table S3	Table S2	results/tables/primary_v1/publication_triage_v1/table_s2_gene_set_ranking.csv
Table S4	Table S3	results/tables/primary_v1/publication_triage_v1/table_s3_discrepancy_review.csv
Table S5	Table C1	results/tables/primary_v1/compounds_v1/table_c1_constituent_inventory.csv
Table S6	Table C2	results/tables/primary_v1/compounds_v1/table_c2_biology_evidence.csv
Table S7	Table C3	results/tables/primary_v1/compounds_v1/table_c3_candidate_ranking.csv (justification columns)
Table S8	Table C4	results/tables/primary_v1/compounds_v1/table_c4_pathway_overlap.csv

Note. Numbers are rounded for display; the source files hold full precision. A small number of columns of the source files (identifiers such as PubChem CIDs and InChIKeys, percentile columns and verbatim quotations) are not reproduced here.

Appendix E Configuration, commands and documentation

Methodological documentation: study protocol docs/protocol.md; architecture and naming docs/architecture.md; data rules docs/data_management.md; figure standards docs/figure_standards.md; compound prioritisation rules docs/compound_protocol.md; robustness plan docs/robustness_plan.md; reviewer matrix docs/agent_workflow.md; phase gates docs/research_roadmap.md; and every decision, open question and investigator ruling in docs/decision_log.md. The page-by-page extraction of the original manuscript is docs/moringa_scientific_linkage.md.

Table S10. Configuration file and script for each step of the pipeline.

Step	Configuration	Script
Preprocessing and quality control	`configs/preprocessing.yaml`	`scripts/run_preprocessing.py`
Primary experiment	`configs/experiment.yaml`	`scripts/run_model.py`
Evaluation	`configs/experiment.yaml`	`scripts/evaluate_models.py`
Explainability	`configs/experiment.yaml`	`scripts/run_explainability.py`
Pathway and cell-line triage	`configs/triage.yaml`	`scripts/generate_triage.py`
Publication tables	`configs/publication.yaml`	`scripts/make_publication_tables.py`
Exploratory M3	`configs/experiment_exploratory_m3.yaml`	`scripts/run_model.py`
Leave-one-tissue-out	`configs/experiment_loto.yaml`	`scripts/run_model.py`
Outcome scales	`configs/experiment_scales.yaml`	`scripts/run_model.py`
Compound evidence and ranking	`configs/compounds.yaml`	`scripts/make_compound_tables.py`
Report figures	`configs/figures_report.yaml`	`scripts/make_report_figures.py`
Negative controls (not executed)	`configs/experiment_controls.yaml`	`scripts/run_model.py`
Coverage simulation (deferred)	`configs/simulation.yaml`	`scripts/run_interval_simulation.py`

Appendix F Reproducibility information

The environment is specified in `environment.yml` (conda environment `explainable-radiosensitivity-ml`, Python 3.11); package versions are recorded in every run record but are not yet pinned in a lockfile. The primary run used six physical CPU cores (twelve logical threads) and about 31 GiB of memory, with bounded multiprocessing and one BLAS thread per worker; the base random seed is 20261021. Tests are in `tests/`; tests that would fit a model are marked `model_fit` and excluded from routine runs. Result folders and figures are write-once, so regeneration requires an explicit version change or an explicit overwrite flag.